SMART INDIA HACKATHON | PROBLEM STATEMENT 26165

RiskRadar

AI/NLP Engine for Serious Injury & Fatality (SIF) Precursor Detection
in Oil India Limited's Unsafe-Act / Unsafe-Condition and Near-Miss Reports

Team NORTHSTAR

Research & Domain Immersion Handbook

About This Handbook

What this is

This is a full domain-immersion handbook for team **NorthStar**, built for your project **RiskRadar** — your response to **Smart India Hackathon Problem Statement 26165**: *"AI/NLP Engine to Detect Serious Injury & Fatality (SIF) Precursors in OIL's Unsafe-Act/Unsafe-Condition and Near-Miss Reports"* for **Oil India Limited (OIL)**.

You don't just need to write code. You need to be able to sit in a judging room and talk fluently with an AI engineer, an HSE manager, an Oil & Gas engineer, and a business evaluator — all in the same five minutes — without sounding like a team that "put a dashboard on top of an LLM." That's what this handbook is for.

How this handbook was built — and its honesty policy

This is not a reformatting of the earlier brainstorming chat you pasted in. Every important factual claim in here — about OIL, about IOGP's Life-Saving Rules, about DEKRA's and EEI's SIF methodology, about competitor products, about real published research — was checked against real, current, public sources (OIL's own annual report and sustainability filings, IOGP's official Life-Saving Rules page, DEKRA's and EEI's own technical papers, vendor press releases, and peer-reviewed literature).

Three rules were followed throughout:

1. **If it's public and verifiable, it's stated plainly, with its source.**
2. **If it's likely internal to OIL and not publicly available, that is said explicitly** — we do not guess at OIL's real incident data, and we never dress up synthetic data as if it were real.
3. **If NorthStar is proposing something (a scoring formula, a data schema, a UI screen) that is not an official OIL/IOGP/DEKRA method, it is labelled as "NorthStar's proposed prototype approach."** A judge should never be able to catch you claiming an invented number as an industry standard.

You will see this phrase a lot: **"Public information could not be verified beyond this."** That is not a weakness in the handbook — it's the model for how you should talk to judges about your own data gaps.

The files in this handbook

#	File	What's inside
00	`00-START-HERE.md`	This file — navigation and how to use the handbook
01	`01-problem-statement-and-oil-india.md`	The PS decoded in plain English; Oil India Limited as a company; how oil & gas actually gets out of the ground and to a refinery
02	`02-hse-organization-and-vocabulary.md`	What HSE/HSSE means; OIL's actual HSE practices (verified); the HSE workflow from observation to trend analysis; a 35-term safety glossary
03	`03-sif-precursors-and-barriers.md`	SIF, SIF Potential, SIF Exposure explained from zero; DEKRA and EEI's real methodologies; safety barriers, Swiss Cheese, Bow-Tie; a hazard reference table
04	`04-iogp-rules-and-hse-processes.md`	All 9 real IOGP Life-Saving Rules; process safety vs personal safety; risk assessment (HIRA/JSA/HAZOP); Permit to Work & LOTO; incident investigation; how to read a safety report
05	`05-riskradar-ai-architecture.md`	RiskRadar's domain model; why NLP beats keywords; NLP/LLM fundamentals; the hybrid AI architecture; a proposed SIF scoring framework; precursor pattern detection
06	`06-data-strategy-and-ai-safety.md`	What OIL data is actually public vs. likely internal; how to design and build a synthetic dataset honestly; how to evaluate the model (precision/recall/F1, why

		accuracy lies); AI failure modes and safeguards
07	`07-competitors-research-regulations.md`	Real competitor products (including one that already does almost exactly this); real peer-reviewed research on this exact problem; India's actual oil & gas safety regulators
08	`08-product-ux-and-killer-feature.md`	The 11-screen dashboard design; the "SIF Precursor Chain" — RiskRadar's core differentiating feature
09	`09-team-execution-and-judge-prep.md`	Role-based learning paths; a build order; ~40 judge Q&A pairs with follow-ups; a red-team attack on your own idea
10	`10-knowledge-check-glossary-references.md`	A certification test with answer key; cheat sheets; the master glossary; full references; the one-page mental model

Suggested reading order by role

- **Everyone, first:** 00 → 01 (Ch.1 only) → the "20 things to remember" and "2-minute explanation" at the end of file 10. That's your shared baseline before anything else.
- **AI/ML members:** 01 → 03 → 04 (Ch.15, 20) → 05 → 06 → 07 (research papers)
- **Backend members:** 01 (Ch.3) → 02 (Ch.6) → 05 (Ch.21) → 06 (Ch.28–29) → 08
- **Frontend members:** 02 (Ch.6–7) → 05 (Ch.21) → 08
- **HSE / domain researcher:** everything in 02, 03, 04 in full — you are the team's safety conscience
- **Product / presentation members:** 01 → 08 → 09 in full

File 09, Chapter 39 has a longer version of these paths.

One page you should bookmark

If you only have ten minutes before you walk into the room, read the **"Final Mental Model"** section at the end of file 10. It's the whole project in one page, plus the 30-second and 2-minute explanations you'll actually say out loud.

PART 01

Part 1–2: The Problem, and The Company

Chapter 1 — What is SIH Problem Statement 26165?

PS in one sentence

> Build an AI/NLP system that reads OIL's everyday safety reports and finds the small number of them that are quietly warning of a possible fatality — even when nobody got hurt.

PS in 30 seconds

Oil India Limited (OIL) collects huge numbers of Unsafe Act (UA), Unsafe Condition (UC), near-miss, and incident reports through its HSSE platform. Right now, these are reviewed manually, in batches, on a schedule (monthly, quarterly). The problem: **a report's actual severity (what happened) and its potential severity (what could have happened) are not the same thing**, and manual, periodic review is not built to catch the pattern of a "quiet but fatal" precursor before it repeats and eventually causes real harm. NorthStar is asked to classify each report as SIF-potential or not, tag it to the relevant IOGP Life-Saving Rule, and surface recurring precursor patterns on a dashboard — so HSE teams can direct attention to where fatal potential is actually concentrated, not just where report volume is highest.

PS in 2 minutes

Every safety-critical industrial company generates two very different kinds of safety information. First, there's what *actually* happened — recorded as injury severity, lost time, medical treatment. Second, there's what *could have* happened if the situation had gone slightly worse — a worker one step closer to a moving load, isolation not verified before a line was opened, a permit skipped under time pressure. The first kind is easy to count. The second kind is buried inside free-text narratives that a human has to read carefully, with domain expertise, to notice.

Global safety research (some of which we verify with real citations in Chapter 10) has shown that the causes of minor injuries and the causes of fatalities are often *not the same causes*. A near-miss with no injury can carry more "fatal potential" than an incident that produced a bandage-level injury. This means that ranking reports by *what happened* — the traditional approach — routinely under-prioritizes the reports that most deserve attention.

The Problem Statement asks NorthStar to build a prototype with three required capabilities:

#	Capability	In plain words
A	**SIF classification**	Read the free text of a UA/UC/near-miss/incident report and predict: does this report show *SIF potential*, or not?
B	**Life-Saving Rule mapping**	If it does show SIF potential, which of IOGP's Life-Saving Rules is it connected to (Energy Isolation, Hot Work, Confined Space, Line of Fire, and so on)?
C	**Pattern discovery**	Across many reports, find recurring combinations of activity + location + barrier failure — and put this on a dashboard that ranks sites and activities by how concentrated the SIF-precursor risk is.

Exact deliverables (from the PS text)

- Ingest OIL's free-text safety reports
- Classify each as SIF-potential vs non-SIF-potential
- Tag each to a relevant IOGP Life-Saving Rule
- Surface recurring precursor patterns (activity, location, barrier failure) via a dashboard
- Expected outcome: a working AI/NLP prototype with an interactive dashboard that ranks sites/activities by SIF-precursor density and auto-maps to Life-Saving Rules

What the solution should NOT attempt to do

- It should not try to predict the exact time or place of a future fatality — that is not what "SIF potential" means, and claiming it will get you torn apart by a technical judge.

- It should not be a new incident-*reporting* app. OIL already has a reporting/HSSE platform; the PS is about the intelligence layer applied to reports that already exist.
- It should not be a computer-vision PPE detector, a blockchain ledger, or an IoT hardware project — none of that is what “AI/NLP Engine” means in this PS.

Common misunderstandings

Wrong interpretation	Why it’s wrong
“We need to build the reporting app itself.”	OIL already has an HSSE platform and reporting workflow (verified in Chapter 2) — you’re building the intelligence layer that reads what’s already collected.
“We predict when a person will die.”	This is not just wrong, it is a dangerous and unsupportable claim — no team should say this to a judge.
“SIF = severe injury, so we just classify by severity.”	This is the exact mistake DEKRA’s research warns against (Chapter 3). SIF *potential* is about what could have happened, not what did.
“More AI is always better, so route everything through one LLM call.”	Judges will ask what happens when the LLM hallucinates a hazard that isn’t there. You need a structured, hybrid answer (Chapter 5, Part 14).

What success looks like to a judge

A convincing demo does not open with an architecture diagram. It opens with **one realistic report going through the whole system**, end to end: text in → hazard/activity/barrier extracted → SIF-potential score with a clear reason → Life-Saving Rule mapped → “this is the Nth similar precursor in the last N days” → a dashboard showing where this is clustering. That single worked example, done convincingly, is worth more than twenty chart types.

> **Judge Tip:** The strongest opening line for your pitch is not “we built an AI system.” It’s: *“OIL’s safety reports already contain the warning signs of the next serious injury — they’re just buried in free text that nobody has time to re-read. RiskRadar surfaces them.”*

Chapter 2 — Oil India Limited (OIL)

Everything below marked with a source is independently verified from OIL's own public materials or reputable secondary sources, current as of mid-2026. Figures that change year to year (revenue, exact headcount, exact safety statistics) are given with their reporting period stated, because a judge may reasonably ask "as of when?"

History and ownership

- OIL's roots trace back to the **1889 discovery of crude oil at Digboi, Assam** — the beginning of India's petroleum industry.
- **Oil India Private Limited was incorporated on 18 February 1959**, formed to develop the newly discovered Naharkatiya and Moran oil fields in Assam.
- In 1961 it became a joint venture between the Government of India and the UK's Burmah Oil Company. In **1981, OIL became a wholly government-owned enterprise**.
- OIL operates under the **Ministry of Petroleum and Natural Gas**, Government of India.
- In **August 2023, OIL was elevated to Maharatna status** — the highest category of Central Public Sector Enterprise (CPSE) in India, making it the 13th Maharatna CPSE. It was previously a Navratna company. Maharatna status gives the company's board significantly greater financial and operational autonomy (equity investments, joint ventures, and M&A up to defined limits without needing case-by-case government approval).
- The Government of India holds a **majority promoter stake (reported around 56–57%)**; the remainder is publicly held (OIL is listed on the NSE and BSE).

What OIL actually does

OIL is a **fully integrated upstream Exploration & Production (E&P) company** — India's oldest, and the **second-largest public-sector oil & gas E&P company after ONGC**. Its core businesses are:

- Exploration, development, and production of **crude oil and natural gas**
- **LPG production**
- **Crude oil and natural gas pipeline transportation** — including the historic 1,157 km Naharkatiya-to-Barauni crude oil trunk pipeline, sometimes called the "crude oil lifeline" of Northeast India
- Expansion into **renewable energy** — captive solar installations, a green hydrogen pilot project (with Himachal Pradesh Power Corporation Limited), Carbon Capture Utilization and Storage (CCUS) exploration, and interest in compressed biogas (CBG) and second-generation ethanol
- Since **March 2021**, OIL is the promoter/holding company of **Numaligarh Refinery Limited (NRL)**, a 3-million-metric-tonnes-per-annum refinery in Assam — adding downstream refining to OIL's upstream E&P base

Where OIL operates

- **Home base:** Assam and Arunachal Pradesh, in India's Northeast — OIL's oldest and largest producing fields.
- **Registered office / field headquarters: Duliajan, Assam.** A separate corporate office operates from Noida, Uttar Pradesh (this split — field HQ in Assam, corporate office near Delhi — is a detail worth knowing, since it tells you OIL's *operational* safety culture is Assam-centered even though a lot of governance sits near Delhi).
- OIL's exploration acreage extends beyond the Northeast into **Rajasthan, Odisha, the Kerala-Konkan coast, and the Andaman offshore area.**
- **International operations:** OIL holds Participating Interest in oil & gas blocks across multiple countries overseas — publicly reported examples include **Russia, the USA, Venezuela, Mozambique, Nigeria, Gabon, Bangladesh, and Libya.** (Older sources also mention Egypt and Yemen; exact current country list can shift with block entries/exits, so if this matters for your pitch, treat it as "several overseas blocks across multiple continents" rather than quoting an exact fixed list.)

Financial scale (context, not the point of your project)

Public financial reporting shows revenue in the tens of thousands of crores of rupees, with results that move with global crude prices — recent public figures put full-year consolidated revenue in the **₹34,000–41,000 crore range** depending on the year, with profit that has varied accordingly. **Do not quote a specific year's number to a judge**

as “current” without checking OIL’s most recent published annual report first — this is exactly the kind of “as of when?” detail that can undercut credibility if it’s stale.

Recognition and strategic direction

- FIPI (Federation of Indian Petroleum Industry) named OIL “Exploration Company of the Year” (2022).
- OIL has articulated a “Mission 4+” ambition (publicly referenced in company materials) around growing annual crude oil and natural gas output, and participates in the government’s **North East Hydrocarbon Vision 2030** initiative to expand the region’s oil & gas economy.

> **Why this matters to RiskRadar:** OIL is not a single facility — it is a Pan-India, increasingly international, increasingly diversified (E&P + refining + pipelines + emerging renewables) company. That means the reports RiskRadar will eventually process could come from very different environments: a decades-old onshore field in Assam, a modern refinery, offshore/coastal exploration, or a solar/green-hydrogen pilot site. A precursor pattern that matters at a mature onshore field (e.g. ageing pipeline integrity) is not the same as one that matters at a refinery (e.g. process safety, loss of containment) or a drilling operation (e.g. well control). Your hazard taxonomy (Chapter 14) needs to be broad enough to span this, and your pitch should show you know OIL is not “just an oil field.”

Chapter 3 — How Oil & Gas Operations Actually Work

You cannot build a credible safety-report classifier without understanding what the people writing those reports are actually doing. This chapter walks the lifecycle a barrel of oil or a cubic meter of gas goes through, stage by stage, and what can go wrong at each one.

```
Exploration → Drilling → Well Completion → Production → Processing
    → Storage → Transportation → Maintenance → Decommissioning
```

Exploration

What happens: Geologists and geophysicists look for hydrocarbon-bearing rock formations underground, using seismic surveys (sending shock waves into the earth and reading the echoes), geological mapping, and exploratory well drilling.

Who works there: Geoscientists, seismic survey crews, exploratory drilling teams — often in remote or difficult terrain.

What can go wrong / major hazards: Vehicle and equipment movement in remote terrain, use of explosive/vibroseis sources for seismic surveys, early well-control risk once exploratory drilling begins.

Typical report language: "Survey crew relocated equipment near an unstable slope…" / "Seismic source truck…"

Drilling

What happens: A drilling rig bores down into the earth to reach the target formation. This is one of the highest-hazard stages in the entire lifecycle.

Who works there: Rig crews, drilling engineers, mud engineers, well-control specialists.

Key equipment: Drill string, rotary table, mud pumps and mud systems, and critically, the **Blowout Preventer (BOP)** — a large valve assembly designed to seal a well and prevent an uncontrolled release ("blowout") if pressure control is lost.

What can go wrong / major hazards: Well "kicks" (an unplanned influx of formation fluid/gas into the wellbore) that can escalate to a blowout if not controlled; suspended loads and dropped objects on the rig floor; rotating/reciprocating machinery; working at height on the derrick; high-pressure mud systems.

Typical SIF scenario: A kick is detected but not immediately shut in per procedure; drilling continues briefly "to finish the stand" — this is a textbook SIF precursor (barrier: well-control procedure; failure: procedural deviation under time pressure).

Well Completion

What happens: Once a productive well is drilled, it is "completed" — casing and cementing are installed, the well is prepared to safely produce, and a wellhead ("Christmas tree" — the valve assembly on top of the well) is installed.

Major hazards: Pressure testing operations, wireline and perforating operations (which can involve explosive charges to perforate the casing), lifting of heavy completion equipment.

Production

What happens: The well flows (or is pumped) to bring crude oil and/or natural gas to the surface, where it passes through wellheads, flowlines, separators (which split oil, gas, and water), and into gathering systems.

Major hazards: Stored pressure in flowlines and separators, H_2S (hydrogen sulfide, a toxic and potentially lethal gas present in some reservoirs) exposure risk, static electricity/ignition risk around hydrocarbon vapor, valve and flange work.

Typical SIF scenario: A valve is believed closed but not verified closed before downstream work begins — an Energy Isolation precursor.

Processing

What happens: Raw production is treated — gas is dehydrated and compressed, oil is stabilized, associated gas may be flared or captured — before it's fit for pipeline transport or refining.

Major hazards: This is where **process safety** (as distinct from personal safety — see Chapter 16) becomes central: loss of containment, overpressure, and fire/explosion risk from large hydrocarbon inventories under pressure and heat.

Storage

What happens: Crude oil, refined products, and LPG are held in tanks, spheres, or bullets before further transport or sale.

Major hazards: Tank overfill, static ignition during tank filling/gauging, vapor space flammability, dyke/bund integrity (the containment berm around a tank that limits spread if the tank fails).

Transportation

What happens: Crude oil and gas move via pipeline (OIL's Naharkatiya–Barauni trunk line is the flagship example), and products move by tanker truck, rail, or (for LPG) cylinder distribution.

Major hazards: Pipeline third-party damage and corrosion-driven loss of containment, road transport incidents (vehicle movement is explicitly one of IOGP's Life-Saving Rules — see Chapter 15), loading/unloading operations at terminals.

Maintenance

What happens: Equipment is inspected, repaired, or overhauled — from routine valve maintenance to major planned "shutdown" or "turnaround" maintenance campaigns that can involve hundreds of contractor workers over days or weeks.

Why this matters most for RiskRadar: Maintenance is disproportionately where **Energy Isolation** and **Line of Fire** SIF precursors concentrate, because it is the activity that most often requires taking "live" equipment out of service, breaking containment, and putting people in direct contact with stored energy that is *supposed* to have been made safe. Several of the worked examples throughout this handbook (and in the original team brainstorming that seeded this project) use a maintenance/isolation scenario for exactly this reason — it is a genuinely representative, high-value example, not an arbitrary one.

Major hazards: Isolation not verified before work starts ("assumed isolated" vs "positively verified isolated" — a critical distinction), line breaking on lines that still hold residual pressure or product, lockout/tagout (LOTO) failures, work at height on elevated equipment, confined space entry into vessels being cleaned or inspected, hot work (welding/grinding) near flammable atmospheres.

Decommissioning

What happens: At end of life, wells are permanently plugged and abandoned, and facilities are dismantled and the site remediated.

Major hazards: Handling of legacy/aged equipment with unknown internal condition, residual hydrocarbon and contamination exposure, demolition hazards.

> **Why This Matters to RiskRadar:** Notice the pattern across every stage — the *specific* equipment changes, but the underlying hazard categories repeat: stored energy, pressure, working at height, line of fire, confined space, hot work. This is exactly why a taxonomy-driven approach (Chapter 21) works better than trying to hand-write rules for every possible piece of equipment. RiskRadar's job is to recognize these *recurring hazard categories* inside free text regardless of which lifecycle stage or piece of equipment the narrative describes.

References for this file

- Oil India Limited, official website (Profile, History, Home, Empowering People pages) — oil-india.com
- Oil India Limited, *Annual Report 2024–25* — oil-india.com/files/financial_results_documents/OIL_India_Annual_Report_2024_25.pdf

- Oil India Limited, *Sustainability Report 2022–23* — oil-india.com/files/sustainability_documents
- Secondary company-profile summaries (financial figures, cross-checked against the above where possible)

PART 02

Part 3–4: HSE, HSSE, and the Language of Safety

Chapter 4 — What is HSE?

HSE stands for **Health, Safety, and Environment**. Some companies (and some of the source material behind this project) use **HSSE** — **Health, Safety, Security, and Environment** — adding “Security” where physical/site security is managed alongside safety (common in oil & gas, where remote sites, pipelines, and valuable product create security as well as safety concerns). OIL’s own public materials primarily use “HSE,” so that’s the term this handbook uses by default, while treating HSSE as a synonym when the source material does.

- **Health:** protecting workers from occupational illness — chemical exposure, noise, ergonomic injury, heat stress.
- **Safety:** preventing injury from acute events — falls, struck-by, caught-in, fire, explosion.
- **Environment:** preventing and managing spills, emissions, and other environmental impact.
- **(Security, in HSSE usage):** protecting people, assets, and operations from deliberate threats — theft, sabotage, unauthorized site access.

Why oil & gas companies need large HSE organizations: the combination of high-energy processes (pressure, stored hydrocarbons), remote/harsh operating environments, heavy equipment, a large contractor workforce, and the catastrophic potential of major accidents (fire, explosion, toxic release) makes safety management a core operational discipline, not an afterthought — it sits alongside production and finance as something the company is judged on.

Chapter 5 — OIL's HSE Practices and Structure

What is publicly verified

OIL's own public materials (its investor pages and recent Annual Report/sustainability disclosures) describe the following, which we can state with confidence:

- **A firm, explicit commitment to HSE** as described in OIL's FY2024–25 Annual Report as "a cornerstone of our corporate values."
- **An incident/near-miss reporting system** — OIL states that "everyone can report to the safety department or safety committee," and that it has "robust auditing systems in place to evaluate the safety performance of our HSE management systems and controls."
- **Risk-assessment practice**: OIL states it uses **Hazard and Operability Study (HAZOP)** and **Quantitative Risk Assessment (QRA)** to assess workplace hazards, and conducts **Job Safety Analysis (JSA)** for significant jobs.
- **Near-miss reporting portals** have been developed specifically to document unsafe acts, unsafe conditions, and near-miss incidents — with awareness programs to encourage reporting, and specific outreach to encourage **contract workers** to report as well.
- **Mock drills** for critical incidents, to train the workforce in emergency response.
- **A named safety initiative called "KAVACH,"** described as aimed at bringing "significant improvement in safety practices" (public detail on KAVACH's specific mechanics was not found beyond this description — treat it as a named internal safety program rather than assuming its exact content).
- **A measured, published safety statistic:** OIL's FY2024–25 Annual Report states the company achieved its **best-ever LTIFR (Lost Time Injury Frequency Rate) of 0.071** at the end of FY2024–25, which it links to "a strong safety culture in E&P operations."
- **An active "HSE Transformation Plan":** OIL's FY2024–25 reporting states that **Phase-I** (covering a perception survey, gap analysis, and competency assessment) has been completed, and the company is now in **Phase-II**, focused on implementing "a robust HSE Management System for enhanced safety and environmental compliance." This is a very useful, current, and specific fact for your pitch: **it tells you OIL is, right now, actively investing in strengthening its HSE Management System** — which is exactly the kind of moment where a credible AI-assisted prioritization tool is a natural complement, not a competing initiative.

> **Note on LTIFR figures:** Different public sources report different LTIFR figures for OIL depending on the exact reporting period and dataset (a separate figure of 0.209 appears in some coverage, which may reflect a different period, a different reporting boundary — e.g. employees vs. employees+contractors — or a different fiscal year). **Do not quote a single LTIFR number to a judge as if it were an eternal fact** — say "OIL's FY2024–25 Annual Report reports its best-ever LTIFR at 0.071" and cite the source, rather than a bare number.

What is NOT publicly verified (be honest about this)

- **An exact organizational chart** of OIL's HSE department (job titles, reporting lines, headcount of the HSE function specifically) was **not found in public sources**. Do not present an invented org chart as OIL's actual structure.
- OIL's **exact incident/near-miss volumes** (how many UA/UC/near-miss reports it receives per year) were not found in the public sources reviewed for this handbook.
- The precise mechanics of the **KAVACH** initiative beyond its stated purpose were not found publicly.

A generic Oil & Gas HSE structure (explicitly labeled as generic, not OIL's confirmed structure)

Large oil & gas companies worldwide typically organize HSE something like this. This is offered so your team has a working mental model to reason with — **label it as generic/industry-typical when you present it, not as verified OIL structure**:

```
Board / Top Management
        |
   Corporate HSE (policy, standards, performance reporting)
        |
   Business-Unit / Asset HSE (E&P, Refining, Pipelines, etc.)
        |
   Field / Installation HSE Manager
        |
   Site Safety Officers
        |
   Supervisors
        |
   Workers and Contractors
```

Alongside the line-management chain, most oil & gas companies run **Safety Committees** (with worker representation) and a **Safety Council** or equivalent senior forum that reviews performance periodically — and OIL's public material specifically confirms it has active **Safety Committees** where HSE concerns are raised and evaluated, consistent with this generic pattern.

Chapter 6 — How an HSE Team Actually Works Day to Day

```
Observation → Report → Review → Classification → Risk Assessment
   → Investigation → Corrective Action → Verification → Closure
   → Trend Analysis
```

Step	What happens	Who does it	Where RiskRadar helps
Observation	A worker, supervisor, or auditor notices an unsafe act, unsafe condition, or near-miss.	Anyone on site — employee or contractor	—
Report	The observation is written up and submitted — via a portal, kiosk, or paper form (OIL specifically mentions reporting kiosks and drop-boxes to make this accessible to contract workers).	The observer	This is RiskRadar's raw material
Review	Someone (safety officer, supervisor) reads the report and decides how urgently it needs attention.	HSE department / safety committee	**RiskRadar's core value: automated triage before human review**
Classification	The report is categorized (near-miss vs. UC vs. UA; sometimes by severity).	HSE department	RiskRadar adds a *SIF-potential* classification layer on top of whatever severity classification already exists
Risk Assessment	If warranted, a more formal risk assessment is done on the underlying hazard.	HSE / engineering	RiskRadar can pre-surface the hazard, activity, and barrier-failure signals to accelerate this
Investigation	For higher-consequence or high-potential events, a structured investigation determines root cause.	Trained investigator(s)	RiskRadar flags candidates for investigation and provides a starting evidence extract
Corrective Action	Actions are assigned to fix the underlying issue (repair, retrain, revise a procedure).	Responsible manager	—
Verification	Someone confirms the corrective action was actually completed and effective.	HSE / supervisor	—
Closure	The report is formally closed out.	HSE department	—
Trend Analysis	Periodically (monthly/quarterly per the PS), patterns across many reports are reviewed.	HSE leadership	**RiskRadar's second core value: continuous, not periodic, pattern discovery**

> **Why This Matters to RiskRadar:** Look at where "periodic" review currently sits — it's really only happening properly at the last step, Trend Analysis, and often only monthly or quarterly (per the PS itself). Everything upstream of that (Review, Classification) is largely manual, one-report-at-a-time triage. RiskRadar's real product position is: **insert automated SIF-aware triage right after Report and before Review**, and make Trend Analysis continuous instead of periodic. That is a precise, defensible answer to "where does your product actually sit in our workflow?" — much stronger than a vague "we help with safety."

Chapter 7 — Safety Vocabulary (Full Glossary)

For each term: full form → simple meaning → real example → why RiskRadar cares.

Hazard

Meaning: Anything with the potential to cause harm — a source of danger, not yet realized. **Example:** A pressurized hydrocarbon line is a hazard whether or not anyone is ever hurt by it. **Why RiskRadar cares:** Identifying the hazard named or implied in a report's free text is the first extraction step in RiskRadar's pipeline (Chapter 21).

Risk

Meaning: The combination of *how likely* a hazard is to cause harm and *how severe* that harm could be. Risk = Likelihood × Consequence. **Example:** Working at height without fall protection is high-likelihood *and* high-consequence — high risk. **Why RiskRadar cares:** SIF potential is really a proxy for "how severe could this get" — the consequence half of risk — regardless of how likely it is to recur.

Consequence

Meaning: The actual or potential outcome if a hazard is realized. **Example:** For an unverified isolation, the consequence could range from "nothing" to "fatal release." **Why RiskRadar cares:** RiskRadar must reason about *potential* consequence, not just recorded consequence.

Exposure

Meaning: The condition of a person being in a position where a hazard could actually reach and harm them. **Example:** A worker standing under a suspended load is "exposed" to that load; a worker 50 meters away is not. **Why RiskRadar cares:** DEKRA's SIF methodology (Chapter 9) is built around *exposure* as the central concept — not injury.

Control

Meaning: Any measure put in place to reduce a hazard's risk — engineering, procedural, or administrative. **Example:** A pressure relief valve is a control. A permit-to-work requirement is a control. **Why RiskRadar cares:** "Control" and "barrier" are near-synonyms in safety science; RiskRadar's job is partly to detect when a control was **absent, bypassed, or ineffective**.

Barrier

Meaning: A specific defense that stands between a hazard and a harmful outcome. (See Chapter 11 for a full treatment.) **Example:** A closed and verified isolation valve is a barrier between stored pressure and a worker opening a line. **Why RiskRadar cares:** "Barrier failure" extraction is arguably RiskRadar's single most valuable output field.

Critical Control

Meaning: Among all the controls for a given major hazard, the ones whose failure would most directly lead to a catastrophic event — the ones an organization monitors most closely. **Example:** For a wellhead, the Blowout Preventer is a critical control; a handrail nearby is a control, but not a *critical* one for that specific major hazard. **Why RiskRadar cares:** Not all barrier failures are equally important — critical-control failures deserve the highest SIF-potential weighting.

Unsafe Act (UA)

Meaning: A behavior by a person that deviates from a safe way of doing a task. **Example:** Climbing over a barrier instead of using the designated access route. **Why RiskRadar cares:** UAs are one of the four report types the PS names explicitly.

Unsafe Condition (UC)

Meaning: A state of the physical environment or equipment that creates hazard, independent of anyone's behavior in the moment. **Example:** A damaged handrail, a leaking valve, an obstructed emergency exit. **Why RiskRadar cares:** Also one of the four PS-named report types; UCs often point directly at a physical barrier that has degraded.

Near Miss

Meaning: An event where something went wrong, or almost went wrong, but no injury or damage actually occurred. **Example:** A worker almost enters a confined space without checking the gas level first, but a supervisor stops the job. **Why RiskRadar cares:** Even though nobody was hurt, the situation may have had serious or fatal potential — this is the exact category the PS is most concerned RiskRadar not under-value.

Incident

Meaning: An unplanned event that did result in some actual harm, damage, or loss — the broad umbrella term. **Example:** A worker suffers a minor cut requiring first aid. **Why RiskRadar cares:** Incidents give RiskRadar actual-outcome data, but as Chapter 9 explains, actual outcome is a poor predictor of SIF potential on its own.

Accident

Meaning: Often used interchangeably with “incident” in casual speech; in formal safety usage, sometimes reserved for incidents that did cause loss (harm, damage), as distinct from near-misses which did not. **Example:** A fall from height that results in injury. **Why RiskRadar cares:** Mostly a terminology note — don’t let inconsistent use of “incident” vs “accident” in source reports confuse your labeling scheme.

SIF (Serious Injury or Fatality)

Meaning: An actual event that resulted in a life-altering injury or a death. **Why RiskRadar cares:** This is the outcome the whole system is ultimately trying to help prevent — but see Chapter 8 for why RiskRadar mostly reasons about *SIF potential*, not actual SIFs.

SIF Potential (SIF-P / SIFp)

Meaning: A characteristic of an incident or near-miss: the situation could reasonably have produced a serious injury or fatality, had circumstances been slightly different — regardless of what actually happened. **Example:** DEKRA’s benchmark question for SIF potential (see Chapter 9): if this exact situation happened dozens or hundreds of times, is it reasonable to conclude that eventually someone would be seriously hurt or killed? **Why RiskRadar cares:** This is the primary classification target of the entire project.

SIF Exposure

Meaning: The state of a person being exposed to a hazard with credible SIF-consequence potential — the “how” behind SIF potential. **Why RiskRadar cares:** Extracting *what* the worker was exposed to (energy type, proximity) is a concrete, extractable NLP target, more tractable than jumping straight to a single SIF-potential label.

High Potential Event (HIPO)

Meaning: An incident or near-miss identified as having high potential for serious consequences — a term used across the industry (including by IOGP, in its original data analysis behind the Life-Saving Rules). **Why RiskRadar cares:** “HIPO” and “SIF-potential” describe closely related — in many programs, functionally overlapping — ideas; different companies use different exact terms for the same concept.

Leading Indicator

Meaning: A proactive safety measurement — something you track *before* harm occurs, that predicts future safety performance. **Example:** Number of safety observations submitted, permit-to-work compliance rate, near-miss reporting rate. **Why RiskRadar cares:** A working RiskRadar precursor-density dashboard is, itself, a new kind of leading indicator.

Lagging Indicator

Meaning: A reactive safety measurement — counts of things that already happened (injuries, fatalities). **Example:** LTIFR, TRIR, fatality count. **Why RiskRadar cares:** Lagging indicators are “insensitive to high-consequence, low-frequency events” — exactly the gap RiskRadar exists to fill.

CAPA (Corrective and Preventive Action)

Meaning: The formal process of fixing the immediate problem (corrective) and preventing recurrence more broadly (preventive) after an incident or audit finding. **Why RiskRadar cares:** RiskRadar's output should feed into, not replace, an organization's CAPA process.

RCA (Root Cause Analysis)

Meaning: A structured method for tracing an incident back past its immediate/obvious cause to the deeper organizational or systemic cause. See Chapter 19 for specific RCA methods. **Why RiskRadar cares:** RiskRadar's barrier-failure extraction is a lightweight, automatic first pass at what a full RCA would eventually formalize.

PTW (Permit to Work)

Meaning: A formal, documented authorization required before certain higher-risk work can begin, confirming the necessary precautions are in place. **Why RiskRadar cares:** Permit-related language ("permit obtained," "permit not issued," "work started before permit signed") is a strong textual signal of barrier status.

LOTO (Lockout/Tagout)

Meaning: The procedure of physically locking an energy-isolating device in the "off"/safe position and tagging it, so it cannot be inadvertently re-energized while someone is working on the equipment. **Why RiskRadar cares:** LOTO language is one of the clearest, most learnable textual signals for the Energy Isolation Life-Saving Rule.

JSA (Job Safety Analysis) / JHA (Job Hazard Analysis)

Meaning: A structured, step-by-step review of a specific job/task to identify hazards and controls before the work begins. JSA and JHA are largely interchangeable terms (JHA is more common in some regions/industries). **Why RiskRadar cares:** A report that references a task with no JSA on file is a signal worth flagging.

HIRA (Hazard Identification and Risk Assessment)

Meaning: A broader, often facility- or process-level exercise (as opposed to a single-task JSA) to systematically identify hazards and assess their risk. **Why RiskRadar cares:** Same general family as JSA — extraction targets for "was a formal risk assessment referenced."

HAZOP (Hazard and Operability Study)

Meaning: A structured, team-based technique (using guide words like "more," "less," "no," "reverse") to systematically identify what could go wrong in a process design and its consequences. **OIL's own public material confirms it uses HAZOP** for risk assessment. **Why RiskRadar cares:** Mostly relevant at the design/engineering stage rather than for classifying individual field reports, but shows the depth of formal risk process OIL already runs, which RiskRadar should complement rather than duplicate.

HAZID (Hazard Identification)

Meaning: An earlier-stage, often less detailed hazard-identification exercise than a full HAZOP — used to build an initial hazard register. **Why RiskRadar cares:** Same family as HAZOP; mostly relevant background, not a direct extraction target from incident narratives.

MOC (Management of Change)

Meaning: A formal process for evaluating and controlling the risk introduced whenever something changes — equipment, procedure, personnel, or organization — before the change is implemented. **Why RiskRadar cares:** A recurring precursor pattern that started appearing right after some known change is a strong, explainable signal for the "emerging pattern" feature (Chapter 27).

SIMOPS (Simultaneous Operations)

Meaning: Two or more potentially conflicting activities happening in the same area at the same time (e.g., crane lifting operations near active drilling). **Why RiskRadar cares:** SIMOPS language in a report is a red flag worth its own tag — combined activities often multiply risk in ways a single-activity classifier could miss.

PPE (Personal Protective Equipment)

Meaning: Equipment worn to protect an individual — hard hats, safety glasses, gloves, respirators, fall-arrest harnesses. **Why RiskRadar cares:** PPE is almost always the *last* line of defense, not the primary barrier — a report that only mentions “PPE was worn” but nothing about upstream controls is not, by itself, evidence of low SIF potential.

LTI (Lost Time Injury)

Meaning: A workplace injury that results in the injured person being unable to work for at least one full day/shift beyond the day of the injury. **Why RiskRadar cares:** The basis for LTIFR — a lagging indicator, contrasted with RiskRadar’s leading, potential-based approach.

LTIFR (Lost Time Injury Frequency Rate)

Meaning: A standardized rate (commonly per 1,000,000 or 200,000 hours worked, depending on convention) measuring how often Lost Time Injuries occur. **OIL reports an LTIFR of 0.071 for FY2024–25** (its best-ever figure), per its Annual Report. **Why RiskRadar cares:** A key example of a lagging indicator OIL already tracks well — RiskRadar’s dashboard should be presented as *complementary* to LTIFR, not a replacement for it.

TRIR (Total Recordable Incident Rate)

Meaning: A standardized rate covering all OSHA-recordable-type injuries (not just lost-time ones), typically per 100 full-time workers per year. **Why RiskRadar cares:** Same family as LTIFR — DEKRA’s research (Chapter 9) specifically shows why TRIR is a poor predictor of SIF risk on its own.

Process Safety

Meaning: The discipline concerned with preventing loss of containment of hazardous materials that could lead to a major event — fire, explosion, or toxic release — as distinct from personal/occupational safety. **Why RiskRadar cares:** See the full treatment in Chapter 16 — RiskRadar must be able to tell process-safety-flavored precursors (e.g. isolation of a hydrocarbon system) from personal-safety-flavored ones (e.g. a fall hazard) since they call for different downstream expertise.

Personal Safety

Meaning: The discipline concerned with preventing injury to individuals from acute, direct hazards — falls, being struck, cuts, and so on. **Why RiskRadar cares:** See Chapter 16.

> **Common Mistake:** Treating “Near Miss” as automatically low priority because “nothing happened.” This is precisely the trap SIF methodology exists to correct — see Chapter 9.

PART 03

Part 5–7: SIF, Precursors, Barriers, and Hazards

Chapter 8 — Serious Injury & Fatality (SIF): The Core Idea

What is a serious injury? What is a fatality?

A **fatality** is a death resulting from a work-related incident. A **serious injury**, in most SIF frameworks, means a permanent impairment, a life-altering condition, or an injury that — without prompt and correct treatment — would lead to death or long-term/permanent harm. DEKRA's own materials (in the testing, inspection, and certification sector) define it this way: *"a permanent impairment or life-altering state, or an injury that if not immediately addressed will lead to death or permanent or long-term impairment."*

Why does SIF deserve its own category, separate from "all injuries"?

Because — and this is the single most important idea in this entire handbook — **the things that cause minor injuries are often not the same things that cause fatalities.**

DEKRA's research (verified — see Chapter 9 for the full citation) found that:

- **Fatality rates across U.S. industry have stayed roughly flat for around 20 years**, even while overall recordable injury rates have declined significantly.
- **About 25% of OSHA-recordable injuries carry realistic SIF exposure potential** — meaning roughly three-quarters of "recordable" incidents, while worth managing, are not actually where fatal risk concentrates.
- **The potential for serious injury is low for the large majority (around 80%) of non-SIF injuries** — for example, DEKRA specifically points to manual lifting: the most common resulting injury is soft-tissue strain, which is very unlikely to escalate to a fatality no matter how many times it recurs.

The consequence of this finding is huge for how a safety program should allocate attention. If you rank reports purely by "how many happened" or "how bad was the actual outcome," you will spend most of your effort on a category of events that was never going to kill anyone — while the rarer, quieter events that *could* have killed someone get the same (or less) attention as a paper cut.

Actual severity vs. potential severity

This is the distinction the whole project turns on:

	Actual severity	Potential severity
Definition	What really happened	What could have happened if circumstances had been slightly worse
Example	A worker trips on a step, catches the handrail, no injury	Had the handrail not been there, or had they been one step further along (near an open floor edge), the same trip could have been a fatal fall
How it's usually measured	Recorded injury classification (first aid, medical treatment, lost time, fatality)	Requires deliberate judgment — asking "if this kept happening, could it eventually be fatal?"

DEKRA offers a useful mental model for judging potential severity: think of rolling dice. Any single roll of "snake eyes" (two ones) is a matter of chance. But **if you rolled the dice over and over, you would eventually roll snake eyes.** The question to ask about any given situation is not "did this cause harm this time?" but **"if this exact situation repeated dozens or hundreds of times, is it reasonable to conclude a serious injury or fatality would eventually occur?"** If the honest answer is yes, the situation has SIF potential — regardless of what actually happened this time.

DEKRA's own formal definition, which is worth learning close to word-for-word: **an incident has SIF potential when a serious injury or fatality is *reasonably* likely to occur if the exposure continues uncontrolled.** The word "reasonably" is deliberate — DEKRA's own materials use the example: *could a paper cut result in a fatality? Technically yes (infection). Is it reasonably likely? No.* SIF potential is not "any conceivable chain of events," it's a *reasonable* judgment about a *credible* consequence.

SIF, SIF Potential, and SIF Exposure — three related but distinct ideas

- **SIF** — an actual, realized serious injury or fatality.
- **SIF Potential (SIFp)** — a property of an incident: it *could have* reasonably produced a SIF, whether or not it actually did.
- **SIF Exposure** — the underlying condition of a person being exposed to a hazard capable of producing a SIF-level consequence; the “raw material” that SIF potential is judged from.

High Potential Events (HIPO)

Many organizations (and IOGP’s own original data analysis behind the Life-Saving Rules — Chapter 15) use the closely related concept of a **High Potential Event (HIPO)**: an incident or near-miss judged, after review, to have had high potential for serious consequences. In practice, “HIPO” and “SIF-potential” describe substantially overlapping ground, though the exact assessment criteria can differ company to company.

> **Judge Tip:** If a judge asks “what makes your SIF-potential label different from just severity?” — this chapter is your answer, word for word: severity measures what happened; SIF potential measures what could have happened, and DEKRA’s own research shows these two things correlate only weakly, which is exactly why a smarter system is needed.

Chapter 9 — SIF Precursors

What is a precursor?

A **precursor**, in general English, is something that comes before and signals what's about to happen. In safety science, a SIF precursor has a precise, useful definition, verified from DEKRA's own process-safety materials:

> **"A SIF precursor is a high-risk situation in which management controls are either absent, ineffective, or not complied with."**

In other words: a precursor is not the accident itself. It's the *condition that made an accident possible* — the missing, broken, or ignored control that, left unaddressed, will eventually align with the wrong moment and produce a real SIF.

The related industry framing (verified from safety-trade-press coverage of EEI's work): **"Precursors are conditions, events, or actions that, if not mitigated, can result in a serious accident. Precursors serve as warnings."** SIF causes are most often linked to violations of an organization's cardinal or life-saving rules — the rules exist specifically because they cover the highest-risk situations, and precursors are what it looks like when one of those rules starts to erode in practice.

What is a SIF precursor *pattern*, and what is "precursor density"?

- A **precursor pattern** is a *recurring* combination of the same kind of precursor — for example, the same barrier failure (isolation not verified) showing up repeatedly during the same activity (maintenance) at the same or multiple locations.
- **"Precursor density"** — the term the PS itself uses — means how concentrated SIF-precursor signals are for a given site, activity, or hazard category, relative to its total report volume. A site with 5 reports where 3 show real SIF potential has much higher precursor density than a site with 500 reports where 3 show SIF potential — even though the raw count is identical. This is exactly the kind of thing a manual, periodic review is bad at noticing, and exactly what RiskRadar's dashboard should be built to surface.

Why near misses are valuable, and why a zero-injury event can still be extremely dangerous

Because a near miss, almost by definition, is an event where the *only* thing that prevented a SIF was luck, timing, or a barrier that happened to still be working. The underlying exposure was real. If you only study events that produced actual injury, you are systematically ignoring the much larger set of events that reveal exactly the same precursor conditions, minus the bad luck.

Ten realistic Oil & Gas SIF-precursor examples

These are illustrative examples written for this handbook to teach the pattern — **they are not real OIL incident data**, and should never be presented as such.

1. **Maintenance began on a hydrocarbon line where isolation was assumed complete but not positively verified** — no injury, but stored pressure/product was present. *(Energy Isolation precursor.)*
2. **A worker was observed standing directly beneath a suspended load during a lifting operation, briefly, while repositioning** — no dropped-object incident occurred. *(Line of Fire / Safe Mechanical Lifting precursor.)*
3. **A confined-space entry began before atmospheric gas testing was completed**, because the crew was behind schedule. *(Confined Space precursor.)*
4. **Hot work (grinding) was performed near a vent stack without first confirming the area was free of flammable vapor.** *(Hot Work precursor.)*
5. **A work-at-height task was carried out using a make-shift platform instead of the approved access equipment**, because the correct equipment wasn't readily available. *(Working at Height precursor.)*
6. **A permit-to-work was signed, but the actual isolation points listed on the permit did not match what was physically isolated in the field.** *(Work Authorisation + Energy Isolation precursor.)*
7. **A driver was observed using a mobile phone briefly while transporting personnel between sites.** *(Driving precursor.)*

8. **An alarm on a pressure vessel had been in a “silenced”/bypassed state for several days pending a spare part, without a documented risk assessment for the interim period.** *(Bypassing Safety Controls precursor.)*
9. **During a shutdown, two crews were working simultaneously in adjoining areas — one doing crane lifts, one doing hot work — without a documented SIMOPS review.** *(Multiple Life-Saving Rules; a SIMOPS precursor.)*
10. **A near-miss report notes that a valve wheel “felt loose” and was reported after use, rather than before** — a possible early sign of equipment degradation on a safety-critical valve. *(Barrier-degradation precursor — not yet tied to a specific Life-Saving Rule, but worth flagging as “equipment integrity” for RiskRadar’s taxonomy.)*

> **Why This Matters to RiskRadar:** Notice none of these ten examples describes an actual injury. Every one of them would likely be filed as “near miss” or, at most, a low-severity UC/UA in a conventional system — and every one of them carries real SIF potential. This is the exact gap the PS is asking NorthStar to close.

Chapter 10 — SIF Methodologies (Verified)

Different organizations have built independent — but philosophically related — frameworks for identifying and managing SIF potential. Here is what each actually says, verified against their own materials, with what NorthStar should and should not claim to be adopting from each.

DEKRA

Who: DEKRA is a global safety and organizational-reliability consultancy; it describes itself as "a leader and originator" of the SIF prevention field, with over 40 years of relevant client work.

What it says: DEKRA's core finding — verified above in Chapter 8 — is that SIF causes differ from the causes of less-severe injuries, and that traditional lagging metrics (TRIR, LTIFR) mislead organizations into treating all incidents as equally informative about fatal risk. DEKRA has published a **"SIF Potential Indicator"** tool: users select from **15 exposure categories**, answer category-specific questions, and the tool determines SIF potential (SIFp) for that incident. DEKRA's guidance stresses two important practical points: (1) if you can't confidently answer the tool's questions, gather more information and come back — don't guess; and (2) incidents are often the result of a *chain* of events, so **analyze the last link in the chain that created the exposure**, not the whole causal history at once.

Relevance to RiskRadar: DEKRA's "reasonably likely if repeated" test (Chapter 8) is the clearest, most defensible plain-English definition of SIF potential available, and NorthStar should adopt this *definition* directly (with attribution). DEKRA's 15-exposure-category structure is also a strong model for how RiskRadar's own hazard/exposure taxonomy (Chapter 21) should be shaped — a bounded, named list of exposure types, not a free-for-all.

What NOT to claim: NorthStar has not licensed or replicated DEKRA's actual SIF Potential Indicator tool or its exact 15-category list (which is DEKRA's proprietary product) — say "our approach is informed by DEKRA's published SIF-potential framework," not "we use DEKRA's methodology."

EEI (Edison Electric Institute) — Safety Classification and Learning (SCL) Model

Who: EEI is a U.S. trade association representing investor-owned electric utility companies — **note this is an electric-power-sector body, not an oil & gas one.** Its SIF work is genuinely excellent and widely referenced across industries, but it was built for and by electric utilities.

What it says: In 2019, an EEI working group (advised by Dr. Matthew Hallowell) created the **Safety Classification and Learning (SCL) Model** — a method for consistently classifying safety incidents and observations, built around **high-energy assessment** and the **identification of direct controls**. The model works by answering a small set of yes/no questions (roughly: was high energy present? was a direct control in place and effective? what was the actual outcome?) to classify an event into one of **seven incident/observation types**, including categories like **PSIF (Potential Serious Injury or Fatality)**, and further distinctions such as **HSIF/LSIF** (higher/lower-likelihood SIF categories) that can be combined into composite metrics. Separately, EEI's precursor work identified **13 precursors** — specific, reasonably detectable events, conditions, or actions that serve as warning signs of a SIF — **specifically for the electric power sector**, developed through a scientific validation process and informed by precursor research from other industries. EEI's associated "precursor analysis protocol" involves observers engaging with field personnel *before* work begins, checking for the presence of known warning signs.

Relevance to RiskRadar: The **high-energy + direct-control** framing is extremely useful and industry-agnostic — "was there significant stored/released energy, and was there a working direct control between that energy and the person?" is a clean, explainable question RiskRadar's scoring logic (Chapter 26) can be built around. The idea of a small, validated, named list of precursors (their 13, tailored to electric utilities) is also a strong model — **NorthStar should build its own, oil-and-gas-specific precursor list**, rather than importing EEI's electric-utility list as-is.

What NOT to claim: Do not present EEI's 13 precursors as if they apply directly to oil & gas — they don't (their precursor list concerns things like proximity to energized conductors, which is meaningful for electric utilities and only loosely analogous to oil & gas hazards). Cite EEI for its *method*, not its specific content, when talking about RiskRadar.

Campbell Institute (National Safety Council)

Who: The Campbell Institute is the research arm of the U.S. National Safety Council, focused on environment, health, safety, and sustainability leadership research.

What it says: The Campbell Institute has published (2018, cited by EEI's own SCL-model documentation) research proposing PSIF classification schemes involving structured lists of SIF exposure categories — part of the same broader wave of research (alongside DEKRA's 2019 work) that moved the field from "severity-based" toward "potential-based" incident classification.

Relevance to RiskRadar: Useful as a *second, independent corroborating source* that the "classify by potential, not just severity" approach is now a broadly accepted direction in safety science, not one consultancy's opinion.

What NOT to claim: Deeper Campbell Institute methodology details beyond this were not directly verified for this handbook — if you want to cite Campbell Institute specifics in a pitch, look up their original 2018 publication directly rather than relying on secondhand summaries.

IOGP — Life-Saving Rules and Process Safety Fundamentals

Who: The International Association of Oil & Gas Producers — the industry body specifically for the oil & gas sector, and the direct source of the Life-Saving Rules the PS explicitly asks NorthStar to map reports to. Full treatment in Chapter 15.

What it says (short version): IOGP's approach is different in kind from DEKRA/EEI's *classification* methodologies — IOGP's Life-Saving Rules are a **behavioral/final-barrier framework**: nine specific, simple actions, derived from real fatal-incident data, that an individual worker has personal control over. IOGP separately maintains **Process Safety Fundamentals**, aimed specifically at the situations most likely to cause *fatal process-safety events* (loss of containment, major fire/explosion) — a complementary, not identical, category to the Life-Saving Rules, which are more focused on personal safety. See Chapter 16 for the personal-safety-vs-process-safety distinction in full.

Relevance to RiskRadar: This is the **mandatory mapping target** the PS explicitly requires (Chapter 15 covers the actual rules). It's also the most oil-and-gas-specific, most directly authoritative source among everything in this chapter.

Where these methodologies differ, and what NorthStar should synthesize

Framework	Sector	Core mechanism	What RiskRadar should borrow
DEKRA	Cross-industry	SIF-potential judgment via exposure category + "reasonably likely if repeated" test	The core *definition* of SIF potential
EEI SCL Model	Electric utilities	High-energy + direct-control decision logic	The *scoring logic pattern* (energy present? control effective?)
Campbell Institute	Cross-industry (NSC)	Potential-based exposure classification	Corroboration that this direction is broadly accepted
IOGP	Oil & gas	9 behavioral Life-Saving Rules + Process Safety Fundamentals	The *mandatory taxonomy* RiskRadar must map to

> **NorthStar Must Know:** No single one of these frameworks is "the" official standard RiskRadar must copy exactly. The strongest, most honest pitch synthesizes DEKRA's *definition*, EEI's *scoring logic pattern*, and IOGP's *oil-and-gas-specific taxonomy* into NorthStar's own proposed prototype method (formalized in Chapter 26) — clearly labeled as NorthStar's synthesis, not any one organization's official product.

Chapter 11 — Safety Barriers

What is a barrier?

A **barrier** is anything that stands between a hazard and a person, preventing the hazard from causing harm. The chain looks like this:

```
Hazard → Threat → Barrier → Exposure (if barrier fails) → Consequence
```

Types of barriers

Type	What it is	Example
Physical / Engineering	Hardware that physically blocks or contains a hazard	A guard rail, a pressure relief valve, a blast wall, a containment dyke
Procedural	A documented process that, if followed, prevents harm	Permit to Work, LOTO procedure, Job Safety Analysis
Human	Depends on a person's competence, verification, or judgment	A worker correctly verifying isolation before starting work
Administrative	Organizational systems that support safe behavior	Training programs, competency assessment, supervision, authorization limits
Preventive	Stops the hazard from becoming an incident at all	Gas detection triggering an automatic shutdown before ignition
Mitigative	Reduces the consequence *after* something has already started to go wrong	Fire suppression systems, emergency shutdown valves, blast-resistant design

Barrier failure — the six ways a barrier stops working

- **Missing** — the barrier was never put in place at all.
- **Failed** — the barrier existed but broke or stopped functioning.
- **Weak** — the barrier exists but wasn't designed adequately for the hazard.
- **Bypassed** — someone deliberately disabled or worked around the barrier (often under time pressure).
- **Degraded** — the barrier still technically exists but has lost some of its effectiveness over time (corrosion, wear, drift).
- **Unverified** — the barrier may or may not be working, but nobody confirmed it before relying on it — this is the single most common pattern in the maintenance-related SIF examples throughout this handbook.

> **Why This Matters to RiskRadar:** "Unverified" is a distinctly different, and arguably more dangerous, category than "failed" — a report that says "isolation was assumed complete" is not describing a broken barrier, it's describing an *unknown* barrier state. RiskRadar's barrier-failure field (Chapter 21) should be able to distinguish these.

Chapter 12 — The Swiss Cheese Model

Developed by psychologist **James Reason**, the Swiss Cheese Model pictures an organization’s defenses as a series of slices of Swiss cheese, stacked one behind another. Each slice is a layer of defense (a barrier, in the language of this handbook); each hole in a slice is a weakness in that particular layer.

```
[Defense 1] [Defense 2] [Defense 3] [Defense 4]
    o           o                       o
        o                   o
              o        o           o
```

Normally, the holes in different slices don’t line up — a weakness in one layer is caught by the next layer. An accident happens on the rare occasion when the holes in *every* slice momentarily align, letting a hazard pass straight through all the way to a person.

Reason distinguishes two kinds of causes:

- **Active failures** — the immediate, visible action that triggered the event (a worker skips a step).
- **Latent conditions** — the underlying, often invisible weaknesses baked into the system long before the event (a poorly designed procedure, understaffing, a maintenance backlog) that made the active failure possible or more likely.

> **Why This Matters to RiskRadar:** This is exactly why blaming “the worker who made a mistake” is usually the wrong lesson from a SIF precursor. A worker skipping isolation verification (an active failure) is far more informative when read alongside *why* — was there a latent condition, like chronic time pressure or an unclear procedure, that made the skip more likely? RiskRadar’s job is mostly to flag active-failure signals in text, but a well-designed RCA workflow downstream (Chapter 19) should always ask about latent conditions too — and your pitch is stronger if you show you understand this distinction rather than implicitly treating every flagged report as “worker error.”

Chapter 13 — Bow-Tie Analysis

A **Bow-Tie diagram** visually connects the *causes* of a hazardous event to its *consequences*, with the barriers on both sides:

```
THREATS  →  [Preventive Barriers]  →  TOP EVENT  →  [Mitigative Barriers]  →  CONSEQUENCES
```

- **Threats** are the things that could cause the top event (e.g., corrosion, operator error, equipment failure).
- **Top Event** is the specific undesired event at the center — e.g., “loss of containment of hydrocarbon from a pipeline.”
- **Preventive barriers** sit on the left — they stop a threat from causing the top event.
- **Mitigative barriers** sit on the right — they reduce how bad the consequence is, *given that* the top event has already happened.

How this connects to RiskRadar’s precursor graph (Chapter 27): A report describing a barrier failure is, in Bow-Tie terms, describing a hole on the *left* side of the diagram — a preventive barrier that didn’t hold. As precursor patterns accumulate across many reports, RiskRadar’s precursor graph is effectively building an evidence-based, continuously updated Bow-Tie for each major hazard — showing which preventive barriers are actually degrading in practice, not just in theory.

Chapter 14 — Major Oil & Gas Hazards

For each hazard: what it is, why it can produce a SIF, typical barriers, typical precursor signals in report text, and its most likely related IOGP Life-Saving Rule (full rule detail in Chapter 15).

Hazard	What it is	Why it can cause SIF	Typical barriers	Typical precursor signals in text	Likely Life-Saving Rule
Stored/pressurized energy	Hydrocarbon or hydraulic pressure held in a line, vessel, or system	Sudden uncontrolled release can strike, burn, or asphyxiate a worker	Isolation valves, pressure relief, verified de-pressurization	"isolation assumed," "residual pressure," "line still connected"	Energy Isolation
Electrical energy	Live circuits, transformers, switchgear	Electrocution, arc flash burns	LOTO, insulated tools, verified de-energization	"circuit not confirmed dead," "worked live"	Energy Isolation
Hydrocarbon release / loss of containment	Uncontrolled escape of oil or gas from its intended containment	Fire, explosion, toxic exposure	Vessel/pipe integrity, gas detection, emergency shutdown	"gas smell reported," "sheen observed," "pressure drop noted"	(Process Safety Fundamentals — see Ch.16)
H_2S / toxic gas	Hydrogen sulfide and similar toxic gases present in some reservoirs/processes	Rapid incapacitation or death at high concentrations	Gas detection, forced ventilation, breathing apparatus	"gas alarm activated," "detector not calibrated," "entered without testing"	Confined Space (if enclosed)
Fire / explosion	Ignition of flammable vapor or liquid	Burns, blast injury, fatality	Ignition-source control, vapor monitoring, fire suppression	"ignition source present," "hot work near vent"	Hot Work
Working at height	Any work above ground/floor level (commonly 1.8m+, see Ch.15)	Falls causing serious injury or death	Fall-arrest harness, guardrails, secured anchor points	"no harness observed," "make-shift platform," "leaning ladder"	Working at Height
Dropped objects / suspended loads	Tools, equipment, or lift loads that could fall	Struck-by injury, often fatal from height	Load securing, exclusion zones, tool tethering	"worker under load," "unsecured tool at height"	Line of Fire / Safe Mechanical Lifting
Line of fire	Being in the path of moving equipment, released energy, or a falling/moving object	Struck-by or caught-in injury	Positioning awareness, exclusion zones, barricading	"stood near swinging load," "between vehicle and wall"	Line of Fire
Confined space	An enclosed or partially enclosed space with restricted entry/exit and potential for hazardous atmosphere	Asphyxiation, toxic exposure, entrapment	Atmospheric testing, entry permit, standby person	"entered before gas test," "no attendant present"	Confined Space
Hot work	Any work producing an ignition source (welding, grinding, cutting)	Fire/explosion if flammable atmosphere present	Fire watch, gas testing, hot-work permit	"grinding near open drain," "permit not renewed"	Hot Work
Vehicle movement	Site traffic, transport	Vehicle collision	Seatbelts, speed	"phone use while	Driving

Vehicle movement / driving	Site traffic, transport between locations	Vehicle collision, struck-by pedestrian	Seatbelts, speed limits, journey management, fit-for-duty driving	“phone use while driving,” “excessive speed noted”	Driving
Lifting operations	Crane, hoist, or manual lifting of loads	Struck-by, crush injury	Rigging inspection, load charts, exclusion zones	“sling condition not checked,” “lift plan not followed”	Safe Mechanical Lifting
Working authorization gaps	Work proceeding without proper authorization/permit alignment	Multiple hazard types can go unmanaged simultaneously	Permit-to-work system, authorization sign-off	“work started before permit signed,” “scope changed without re-authorization”	Work Authorisation
Bypassed safety controls	An alarm, interlock, or safety device deliberately disabled or silenced	Removes the very control meant to prevent escalation	Change-management/risk assessment before any bypass, time-limited bypass with compensating measures	“alarm silenced,” “interlock jumpered,” “bypass not logged”	Bypassing Safety Controls
Excavation	Digging that could disturb buried services or collapse	Trench collapse, utility strike	Permit, shoring, utility locating	“trench unshored,” “dig without clearance”	(Largely folded into Work Authorisation since the 2017/18 IOGP revision — see Ch.15)
Process upset / blowout	Loss of well or process control	Uncontrolled release, potential catastrophic event	Well-control procedures, blowout preventer, process interlocks	“kick observed,” “pressure anomaly not actioned”	(Process Safety Fundamentals)

Judge Tip: If asked “how do you decide which Life-Saving Rule applies when a report seems to touch two or three?” — the honest, strong answer is: RiskRadar should support **multiple tags per report** rather than forcing a single label, because real precursors often do span more than one rule (see example #9 in Chapter 9, the SIMOPS example, which touches both Safe Mechanical Lifting and Hot Work at once).

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PART 04

Part 8–11: IOGP’s Rules, Process Safety, and the Mechanics of HSE

Chapter 15 — IOGP Life-Saving Rules (Verified, In Full)

Everything in this chapter is verified directly from IOGP's own official Life-Saving Rules page and its published FAQ (iogp.org). This is the single most judge-relevant chapter in the whole handbook, because Life-Saving Rule mapping is an explicit, named deliverable in the PS.

Where the Rules came from

- IOGP originally published Life-Saving Rules starting around **2010**, based on analysis of **1,484 fatal incidents (1991–2010)** and **1,173 High Potential Events (2000–2010)** reported to IOGP. IOGP's members concluded that following these rules could have prevented roughly **70% of those fatalities**.
- That original set had **18 rules (8 "Core" + 10 "Supplemental")**.
- In **2017–2018**, following analysis of a further ten years of fatal-incident data, IOGP published **Report 459**, simplifying the set down to **9 rules**, rewritten from the worker's own first-person perspective ("I…" statements) for clarity.
- IOGP states that **in the ten years covered by this analysis (2008–2017), 376 people lost their lives** in incidents that might have been prevented by following one of the Rules.
- The content of the Rules was checked against safety data from other recognized bodies, including **ARPEL, CONCAWE, NIOSH, and OSHA**, to confirm relevance across the industry, not just IOGP's own member companies.
- IOGP is explicit that the Rules are **not intended to cover every possible hazard.** They exist to draw attention to the specific activities data shows are most likely to lead to a fatality, and to the simple, individually-controllable actions that prevent it. They are designed to work *alongside* — not instead of — a company's full management system, competent people, and site procedures. IOGP describes the Rules as **"a final barrier"** — the layer of defense an individual worker has complete personal control over, for use when every other layer has already failed.

The Nine Life-Saving Rules

Each rule pairs a short set of specific, observable actions ("I will…" / "I will never…") with the hazard it targets.

1. Bypassing Safety Controls *Obtain authorisation before overriding or disabling safety controls.* Covers deliberately disabling or ignoring an alarm, interlock, or other safety device without going through a proper risk assessment and authorization first. **Example report language:** "Pressure alarm was silenced pending spare part delivery, without documented risk assessment." **Why it exists:** Removing a control that exists specifically to prevent escalation is one of the fastest ways to remove the very thing standing between a hazard and a person.

2. Confined Space *Obtain authorisation before entering a confined space.* Covers entry into any enclosed or partially enclosed space with restricted entry/exit and a potential for hazardous atmosphere. **Example report language:** "Entry made into vessel before gas testing was completed." **Why it exists:** Confined spaces can develop lethal atmospheres (oxygen deficiency, toxic or flammable gas) invisibly and rapidly.

3. Driving *Follow safe driving rules.* IOGP's own published "I" statements: **always wear a seatbelt; do not exceed the speed limit and reduce speed for road conditions; do not use phones or operate devices while driving; be fit, rested, and fully alert while driving; follow journey management requirements. Example report language:** "Driver observed using mobile phone briefly during personnel transport." **Why it exists:** Road transport is one of the largest causes of workplace fatality across many industries, including oil & gas, because of the sheer volume of driving involved in remote/dispersed operations.

4. Energy Isolation *Verify isolation and zero energy before work begins.* Covers identifying, isolating, and/or locking and tagging out all relevant energy sources (electrical, mechanical, hydraulic, pneumatic, stored pressure) before work starts. **Example report language:** "Isolation was assumed complete but positive verification was not performed before line breaking." **Why it exists:** Working on equipment that is still energized — or thought to be isolated but isn't — is one of the clearest, most repeated patterns behind fatal incidents in maintenance work (see Chapter 9's worked examples).

5. Hot Work *Control flammables and ignition sources.* Covers identifying and controlling ignition sources before starting any task that could generate one — welding, grinding, cutting — including confirming the area is free of flammable material and vapor, and obtaining authorization. **Example report language:** "Grinding work performed near open drain without confirming vapor-free atmosphere.” **Why it exists:** In an industry built around flammable hydrocarbons, an uncontrolled ignition source is a direct path to fire or explosion.

6. Line of Fire *Position yourself to avoid the line of fire.* This rule is philosophically different from the others — it is about active, ongoing awareness of your position relative to moving equipment, released energy, suspended loads, and other people’s activity, rather than a single one-time check. IOGP’s own materials note that **a majority of member-reported fatalities may have been prevented had the individual simply not been in the line of fire** at the critical moment. IOGP has published newly expanded (2025) supporting materials identifying **10 specific Line of Fire hazard categories** (drawn from Fatality and Permanent Impairment data analysis) to help workers recognize this hazard in the moment. **Example report language:** “Worker briefly positioned beneath suspended load while repositioning rigging.” **Why it exists:** Struck-by and caught-in-between events are consistently among the top causes of fatality in industrial settings.

7. Safe Mechanical Lifting *Plan lifting operations and control the area.* Covers proper planning of crane and rigging operations, verified equipment condition, and control of the area beneath and around a lift. **Example report language:** “Rigging inspection not documented prior to lift.” **Why it exists:** A dropped or swinging load can deliver enormous, often fatal, force with very little warning.

8. Work Authorisation *Work with a valid permit when required.* Covers ensuring that higher-risk work only proceeds once it is properly authorized — the permit-to-work system. IOGP’s own FAQ material notes this rule has effectively absorbed much of what a separate “excavation” rule used to cover (excavation-related fatalities were about 2% of the 2008–2017 dataset, not enough on its own to justify a dedicated rule, but the underlying authorization failure pattern is captured here). **Example report language:** “Work commenced on the isolated section before the permit was formally signed.” **Why it exists:** A permit isn’t paperwork for its own sake — it is the moment where all the other rules are supposed to be actively checked before work starts.

9. Working at Height *Protect yourself against a fall when working at height.* Covers using proper fall protection when working in an elevated or otherwise unprotected area. IOGP recommends **1.8 metres (about 71 inches)** as the height threshold above which fall protection should be worn as a rule of thumb — while explicitly deferring to local legal requirements where those differ (use the stricter of the two). **Example report language:** “Access to elevated platform made via unsecured ladder without fall-arrest equipment.” **Why it exists:** Falls from height are one of the most common causes of fatality across all industrial sectors worldwide.

Start Work Checks

Alongside the nine Rules, IOGP publishes **Start Work Checks** — a complementary tool that helps frontline workers, right before beginning a task, actively confirm the specific controls and safeguards relevant to that task are actually in place. Where the Life-Saving Rules state *what* matters, Start Work Checks are a practical *how* — a short verification ritual immediately before work begins.

“Stop Work” and “Check and Reassess”

Consistent with the Rules being a *personal, final* barrier, IOGP’s guidance strongly reinforces that any worker who cannot confirm a Rule can be followed (for example: a vehicle with no functioning seatbelt) should **stop the job, notify a supervisor, and carry out a risk assessment** before proceeding — rather than continuing and hoping for the best. IOGP is also explicit that **a Life-Saving Rule violation remains a violation even if, on investigation, the specific case could not actually have led to a fatality** — the rule was still broken, and the reasons for that should be understood and addressed, not dismissed because “nothing happened this time.”

Relationship to Process Safety Fundamentals

IOGP separately publishes **Process Safety Fundamentals** — a complementary set of guidance aimed specifically at the situations most likely to cause a *fatal process-safety event* (a major loss of containment, fire, or explosion), as distinct from the Life-Saving Rules’ focus on personal, individually-controllable safety actions. See Chapter 16 for the full distinction.

Are the Rules relevant to contractors?

Yes, explicitly. IOGP’s own FAQ notes contractors carry out roughly **80% of the work** in the oil & gas industry, and one motivation for standardizing on a single, industry-wide set of Rules (rather than every operator having its own “golden rules”) is so contractors moving between different companies aren’t forced to learn multiple different versions of the same safety logic — which itself introduces risk, since workers can get confused about which company's rules apply where.

> **NorthStar Must Know:** IOGP's Rules are explicitly **not exhaustive of every industrial hazard** (excavation and drug/alcohol rules, for instance, were deliberately folded into other rules or left as separate site-level policy in the 2018 simplification). This means RiskRadar's hazard taxonomy (Chapter 21) should be broader than "only the 9 Rules" — a report can carry real SIF potential without cleanly mapping to any single Life-Saving Rule, and your system needs an honest way to say so rather than forcing a bad-fit tag.

Chapter 16 — Process Safety vs. Personal Safety

This is a distinction judges specifically listen for, because conflating the two is a common and telling mistake in student pitches.

Personal safety

Definition: Preventing injury to an individual from a direct, acute hazard. **Example:** A worker falls from height. **Typical scale of consequence:** Usually affects the individual(s) directly involved.

Process safety

Definition: Preventing the loss of containment of a hazardous material (or the loss of control of a hazardous process) in a way that could cause a major event. **Example:** A hydrocarbon containment failure leads to a vapor cloud that ignites, causing a fire or explosion. **Typical scale of consequence:** Can affect many people, not just the person directly involved — including people not even part of the original task.

Key related terms

- **Process Safety Management (PSM):** the overall organizational discipline and set of systems (hazard analysis, mechanical integrity, management of change, and more) built to prevent major process-safety events.
- **Loss of Containment:** any unplanned release of a hazardous material from its intended boundary (a pipe, a vessel, a tank).
- **Major Accident Hazard:** a hazard with the potential to cause a major accident — typically defined as an event with the potential for multiple fatalities, severe environmental damage, or major asset loss.
- **Critical controls:** as defined in Chapter 11 — for process safety specifically, these are the barriers (relief systems, shutdown systems, containment integrity) that most directly prevent a major event.

How IOGP frames the relationship

IOGP's own materials describe the **Life-Saving Rules** and **Process Safety Fundamentals** as complementary, not identical, frameworks: the Life-Saving Rules are built around individually-controllable *personal safety* actions, while Process Safety Fundamentals are built to draw attention to the situations most likely to cause a fatal *process safety event*. A worker can follow every Life-Saving Rule perfectly and still be caught in a major process-safety event caused by a systemic containment or design failure unrelated to their own individual actions that day.

How RiskRadar should handle both, without confusing them

A report about a worker climbing without fall protection is a **personal-safety / Life-Saving Rule** case. A report about a valve found leaking, or an alarm that was silenced on a pressure vessel, is a **process-safety** case — and while it may still map to a Life-Saving Rule (Bypassing Safety Controls, Energy Isolation), it deserves an additional "process safety relevant" flag, because these events can escalate to a scale that a purely personal-safety framing would understate. **RiskRadar's domain model (Chapter 21) should carry a separate `process_safety_relevant` flag alongside the Life-Saving Rule tag**, rather than treating "mapped to a rule" as the only signal that matters.

Chapter 17 — Risk Assessment

The core building blocks

- **Hazard identification** — systematically finding what could go wrong, before deciding how likely or severe it is.
- **Likelihood** — how probable a hazard is to actually cause harm.
- **Consequence** — how bad the harm would be if it did occur.
- **Risk matrix** — a grid (commonly likelihood on one axis, consequence on the other) used to categorize risk as low/medium/high/extreme, guiding what level of control and approval is required.
- **Residual risk** — the risk that remains *after* controls are applied; no control reduces risk to zero.
- **ALARP (As Low As Reasonably Practicable)** — the principle that risk should be reduced until the cost/effort of further reduction is grossly disproportionate to the benefit gained — a common legal/regulatory standard in many jurisdictions for what counts as "safe enough."

The named methods

- **HIRA (Hazard Identification and Risk Assessment):** a broad, systematic risk-assessment process, often facility- or activity-level.
- **JSA / JHA (Job Safety Analysis / Job Hazard Analysis):** a task-level, step-by-step breakdown of a specific job to find hazards and controls before work starts. **OIL's own public materials confirm it conducts JSA for significant jobs.**
- **TRA (Task Risk Assessment):** similar in spirit to JSA — a task-focused risk review, terminology varies by company.
- **HAZID (Hazard Identification):** an earlier-stage, broader hazard-identification workshop, often feeding into a later HAZOP.
- **HAZOP (Hazard and Operability Study):** a rigorous, team-based, guide-word-driven ("more," "less," "no," "reverse," and so on) review of a process design to systematically find deviations and their consequences. **OIL's own public materials confirm it uses HAZOP.**

Chapter 18 — Permit to Work & Job Controls

Permit to Work (PTW)

A **Permit to Work** is a formal, documented authorization confirming that specific precautions have been verified before a higher-risk task can begin. It typically names the task, the hazards, the required isolations, the people authorized, and a validity period. **OIL's own public disclosures confirm the use of an online work-permit system.**

Lockout/Tagout (LOTO) and Isolation

LOTO is the physical practice of locking an energy-isolating device (a valve, a circuit breaker) in the safe position and tagging it, so it cannot be inadvertently operated while someone is working on the equipment downstream of it. **Isolation** is the broader concept — LOTO is one specific, physical way of achieving verified isolation.

The critical distinction, repeated throughout this handbook because it is genuinely the single most common SIF-precursor pattern in maintenance work: **"isolation completed" is not the same claim as "isolation verified."** The first describes an intended state; the second describes a confirmed, checked state. RiskRadar's extraction logic (Chapter 21) should specifically try to distinguish these two very different-sounding but textually similar phrases.

Gas Testing

Before entry into a confined space (or before hot work in a potentially flammable atmosphere), the air is tested for oxygen level, flammable gas concentration, and toxic gas presence, using a calibrated gas detector. A report mentioning entry or work *before* gas testing was completed is a strong, learnable Confined Space or Hot Work precursor signal.

Confined-Space, Hot-Work, and Excavation Permits

These are specialized permit types layered on top of the general PTW system, each with its own specific pre-work checklist (gas testing and standby person for confined space; ignition-source control and fire watch for hot work; utility clearance and shoring assessment for excavation).

Work-at-Height Controls

Fall-arrest harnesses connected to a verified anchor point, guardrails around elevated platforms and floor openings, and — per IOGP's guidance in Chapter 15 — fall protection as standard practice above roughly 1.8 metres.

> **Why This Matters to RiskRadar:** Every one of these systems (PTW, LOTO, gas testing, permits) is itself a *barrier* in the Chapter 11 sense. A failure inside any of them is, definitionally, a precursor. This is why RiskRadar's precursor taxonomy should be organized around these named control systems rather than being a flat list of hazards — it makes the "which barrier failed" extraction target concrete and learnable.

Chapter 19 — Incident Investigation

The general lifecycle

```
Incident → Secure the scene → Collect evidence → Interview →
Build a timeline → Identify immediate causes → Identify underlying/root
causes → Identify which barriers failed → Corrective action → Verification
```

Named methods (brief, conceptual overview only — enough to discuss intelligently, not to run one yourself)

- **5 Whys:** repeatedly asking “why did that happen?” about each answer, until you reach a root cause rather than stopping at the first, most obvious explanation.
- **Fishbone (Ishikawa) Diagram:** a visual technique organizing potential causes into categories (people, equipment, procedure, environment, and so on) branching off a central “spine.”
- **Fault Tree Analysis (FTA):** works backward from an undesired top event, mapping the logical combinations of failures that could produce it.
- **Event Tree Analysis (ETA):** works forward from an initiating event, mapping the different possible sequences and outcomes depending on which barriers hold or fail.
- **Bow-Tie:** see Chapter 13 — combines threat-side (Fault-Tree-like) and consequence-side (Event-Tree-like) thinking into one diagram.
- **ICAM (Incident Cause Analysis Method):** a structured investigation methodology widely used in resource and energy industries, distinguishing absent/failed defenses, individual/team actions, task/environmental conditions, and organizational factors.
- **TapRooT®:** a proprietary, widely licensed root-cause-analysis system and software tool used across many industrial sectors.
- **Tripod Beta:** a root-cause methodology (originally developed in the oil & gas sector specifically) built around identifying underlying organizational “failure states” behind observed unsafe acts and conditions.

Where AI can help without replacing investigators: RiskRadar’s automatic extraction (hazard, activity, barrier, exposure) can hand a human investigator a fast, evidence-linked starting point — but the actual root-cause judgment, especially distinguishing active failures from latent organizational conditions (Chapter 12), remains a human, expert task. This is a good, honest line for judge Q&A.

Chapter 20 — Understanding OIL Safety Reports

A realistic example report

(This is an illustrative example written for this handbook, not a real OIL report — never present anything like this to a judge as actual OIL data.)

> **Report type:** Near Miss **Date/Location:** [field location], Maintenance activity **Narrative:** “During scheduled maintenance on a hydrocarbon transfer line, work began after the upstream valve was closed. Positive isolation was not verified with a pressure test before flange breaking commenced. Residual pressure was present in the line. The worker positioned near the flange noticed a slight release and immediately stepped back. Work was stopped and the line was re-isolated and verified before continuing. No injury occurred.”

How an HSE expert reads this

An experienced HSE reviewer immediately notices: *isolation was assumed, not verified*; *the worker was in a position of direct exposure at the moment of the release*; *the outcome (no injury) was fortunate timing, not the result of an effective barrier*. A skilled reviewer classifies this as high-priority for follow-up regardless of the “no injury” outcome — but that judgment depends on catching the specific phrase “not verified” and connecting it to the consequence, which requires both attention and domain knowledge, applied consistently across potentially hundreds of similar reports.

How RiskRadar should read this

```
Report text
    |
    ▼
Activity: Maintenance / line breaking
    |
    ▼
Hazard: Stored/residual pressure (hydrocarbon)
    |
    ▼
Exposure: Worker positioned near flange at moment of release
    |
    ▼
Barrier: Positive isolation
    |
    ▼
Barrier Failure: Verification not performed before work began
    |
    ▼
Potential Consequence: Uncontrolled release → fire, chemical exposure,
                        or line-of-fire injury, up to fatal
    |
    ▼
SIF Potential: HIGH (reasonably likely if this exact situation
                repeated — see the DEKRA test in Chapter 8)
    |
    ▼
Life-Saving Rule: Energy Isolation
    |
    ▼
Risk Priority: Escalate for HSE review
```

This is the same worked pattern used throughout this handbook (Chapters 3, 8, 9), because it is genuinely the most representative, highest-value example RiskRadar can be built and demonstrated around — not a coincidence of the earlier team brainstorming that fed into this project.

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PART 05

Part 12–16: RiskRadar’s Domain Model and AI Architecture

Chapter 21 — What Exactly Should RiskRadar Understand?

Before writing any model code, define the exact structured fields RiskRadar is trying to extract from each report. This is your domain ontology — the shared vocabulary your AI/ML, backend, and frontend members all build against.

```
Report
 ├── Report Type         (UA / UC / Near Miss / Incident)
 ├── Activity             (what task was being performed)
 ├── Location             (site / installation, if available)
 ├── Hazard               (from the Chapter 14 taxonomy)
 ├── Energy Type           (stored pressure, electrical, mechanical, gravitational/height, chemical, thermal)
 ├── Exposure              (was a person actually in a position to be harmed? how?)
 ├── Barrier                (which control was supposed to prevent harm)
 ├── Barrier Failure Type   (missing / failed / weak / bypassed / degraded / unverified — Chapter 11)
 ├── Potential Consequence  (a short, plain-English description — not a fixed enum, since real consequences vary
too much to force into a short list)
 ├── SIF Potential          (High / Medium / Low, with a confidence score — Chapter 26)
 ├── Life-Saving Rule(s)    (zero, one, or more of the 9 rules — Chapter 15)
 ├── Process Safety Relevant (boolean flag — Chapter 16)
 ├── Confidence             (how sure the model is about the extraction and score)
 └── Recommended HSE Action  (e.g. "escalate for review," "log only," "route to investigation")
```

Every field above should be traceable back to a specific piece of the source text — this “evidence linking” is what makes RiskRadar’s output explainable (Chapter 25) rather than a black-box score.

Chapter 22 — Why AI/NLP Is Useful Here

The core problem: the same word can mean opposite things for safety

> **“Isolation completed.”** vs. **“Isolation was not verified.”**

A simple keyword search for “isolation” would treat both sentences identically — both contain the word. But the first describes a barrier that (at least on its face) is intact; the second describes a barrier whose state is genuinely unknown, which — per Chapter 8’s DEKRA-verified logic — is exactly the kind of situation with real SIF potential. **Keyword matching cannot tell these apart. Understanding the surrounding context — the negation, the specific verb, what comes before and after — requires semantic understanding of the sentence, not just presence/absence of a word.**

This is the fundamental reason the PS asks for an AI/NLP engine rather than a rules-based keyword search: **context determines meaning**, and free-text safety narratives are full of exactly this kind of subtle, safety-critical distinction.

Chapter 23 — NLP Fundamentals (No Unnecessary Math)

- **Tokenization:** breaking text into smaller pieces (words, sub-words) a model can process.
- **Embeddings:** representing a word, sentence, or document as a list of numbers (a "vector") positioned in a mathematical space such that texts with similar *meaning* end up near each other — even if they don't share exact words. This is how a model can recognize "loss of balance from a high platform" and "fell from height" as related, without either containing the same keyword.
- **Semantic similarity:** measuring how close two pieces of text are in that embedding space — the mechanism behind "find similar reports."
- **NER (Named Entity Recognition):** automatically finding and labeling specific things in text — a location, an equipment type, a date.
- **Classification:** assigning a category (or categories) to a piece of text — this is the core mechanism behind SIF-potential labeling and Life-Saving Rule mapping.
- **Clustering:** automatically grouping similar items together *without* pre-defined categories — useful for discovering precursor patterns you didn't already know to look for.
- **Transformers:** the modern neural-network architecture behind almost all current high-performing NLP models — built around an "attention" mechanism that lets the model weigh which words in a sentence matter most for understanding any given word's meaning in context.
- **BERT (Bidirectional Encoder Representations from Transformers):** a widely used transformer model, specifically good at *understanding* text (as opposed to *generating* new text) — a strong fit for classification tasks like SIF-potential labeling. Several real published studies on exactly this kind of safety-report classification use BERT or BERT-family models (verified — see Chapter 34 for citations).
- **Sentence embeddings:** embeddings computed for a whole sentence or short passage at once (rather than word by word), well suited to comparing whole report narratives against each other for the "find similar precursors" feature.

Chapter 24 — LLMs (Large Language Models)

- **LLMs** are large transformer-based models trained to understand and generate natural language, and — increasingly — to follow instructions and reason over provided context.
- **Prompts:** the instructions and context given to an LLM to produce a desired output.
- **Structured output:** instructing an LLM to respond in a specific, machine-readable format (like JSON matching the domain model in Chapter 21) rather than free-form prose — essential for RiskRadar, since downstream code needs to reliably parse the model's output.
- **Function calling / tool use:** letting an LLM call external functions or tools as part of producing its answer — potentially useful for RiskRadar to look up, e.g., a site's report history while reasoning about a new report.
- **RAG (Retrieval-Augmented Generation):** giving an LLM relevant reference material (e.g., the actual text of the IOGP Life-Saving Rules, or a set of past similar reports) at the moment it's answering, rather than relying purely on what it happened to learn during training. This is a strong fit for RiskRadar's Life-Saving Rule mapping step, since it grounds the model's tagging in the actual rule definitions rather than the model's possibly-imprecise memory of them.
- **Vector databases:** specialized databases for storing and quickly searching embeddings — the infrastructure behind both RAG and "find similar past reports."
- **Hallucination:** when a model generates plausible-sounding but false or unsupported content — for RiskRadar, this could mean inventing a hazard, barrier, or consequence that the source text doesn't actually support. This is the single biggest reason RiskRadar cannot be "just an LLM call" (Chapter 25).
- **Confidence / context windows / temperature:** technical knobs governing how much text a model can consider at once, and how deterministic vs. varied its output is — worth knowing the terms exist, without needing deep mastery for a hackathon prototype.

Where an LLM is useful, and where it should not be trusted alone: useful for semantic extraction — pulling structured fields (hazard, activity, barrier) out of messy free text, and for RAG-grounded rule mapping. **Not to be trusted alone** for the final SIF-potential decision, because that decision has real consequences if wrong, and an LLM's confident-sounding output is not the same thing as a *justified* output.

Chapter 25 — Why RiskRadar Should Not Be “Just GPT”

The naive (and fragile) approach

```
Report → LLM → "SIF: Yes/No"
```

This is fragile because: it gives no evidence for its answer (not explainable — a real problem in a safety-critical application), it can hallucinate a hazard that isn’t in the text, its behavior can shift unpredictably between similar-sounding reports, and — most importantly for a judge conversation — **you cannot audit or improve it in any structured way** when it gets something wrong.

The hybrid approach NorthStar should build instead

```
Report
   |
   ▼
AI Extraction (LLM/NLP layer — pulls structured fields per Chapter 21's ontology)
   |
   ▼
Domain Taxonomy (maps extracted terms to the fixed hazard/barrier/Life-Saving Rule vocabulary from Chapters 14–
15)
   |
   ▼
Rule Engine / Scoring Logic (deterministic scoring based on the extracted, taxonomy-mapped fields — Chapter 26)
   |
   ▼
SIF Assessment + Confidence + Evidence
   |
   ▼
Human Review (especially for high-severity or low-confidence cases)
```

The key architectural idea: **the LLM’s job is narrow — extract evidence from text into a structured shape. The scoring/classification logic is a separate, deterministic (or at least auditable) layer that operates on that structured evidence, not on raw model “vibes.”** This means:

- If the score is wrong, you can inspect *which extracted field* drove it, and fix that layer specifically.
- The system can show its work: “flagged HIGH because: energy type = stored pressure; barrier = isolation; barrier failure = unverified; exposure = worker present at flange.”
- Confidence thresholds (Chapter 31) can route uncertain cases to a human, rather than forcing a single automated answer on every report.

> **Judge Tip:** When asked “what happens if the LLM hallucinates?” — this is your answer: *“The LLM only performs extraction, not the final safety decision. Extracted evidence is checked against our domain taxonomy, and the SIF score is computed by an auditable scoring layer on top of that evidence — so a hallucinated hazard that doesn’t map to our taxonomy simply doesn’t produce a score, rather than silently producing a wrong one.”*

Chapter 26 — How Should RiskRadar Determine SIF Potential?

Signals informed by the verified methodologies in Chapter 10

- **Exposure** — was a person actually positioned where a hazard could reach them? (DEKRA, EEI)
- **Energy** — was meaningful stored, electrical, mechanical, or other significant energy present? (EEI's SCL Model)
- **Barrier / direct control** — was there a relevant control, and was it present, verified, and effective? (EEI's SCL Model; Chapter 11)
- **Proximity** — how close, in space and time, was the person to the point of potential release/failure?
- **Activity criticality** — is this an activity statistically associated with higher fatal potential (maintenance, confined space entry, work at height, hot work)?
- **Consequence severity if realized** — using the DEKRA "reasonably likely if repeated" test from Chapter 8, not the actual recorded outcome.

A proposed prototype scoring framework — labeled explicitly as NorthStar's own, not an official DEKRA/EEI/IOGP formula

> ⚠ **This is NorthStar's proposed prototype approach, not an industry-standard formula. Never present this to a judge as "the" official SIF scoring method — present it as: "our synthesis of the published methodologies in Chapter 10, adapted for a hackathon prototype."**

A simple, explainable, weighted-signal approach (illustrative starting point — tune with real or synthetic data during development):

```
SIF Risk Score =
      w1 × Exposure Present (0/1)
    + w2 × Energy Level (0–3: none/low/moderate/high)
    + w3 × Barrier Status (0: verified-intact, 1: degraded, 2: unverified, 3: failed/bypassed)
    + w4 × Proximity (0–2: distant/nearby/direct contact)
    + w5 × Activity Criticality (0–2, from a lookup table keyed to the Chapter 14 hazard list)
```

Convert the resulting score into three bands (High / Medium / Low), and always report a **confidence score** alongside the band — driven by how much of the text was clearly extractable vs. ambiguous or missing.

Thresholds, uncertain cases, and escalation

- Reports scoring **High** with **high confidence** → straight to the HSE Priority Queue (Chapter 37).
- Reports scoring **High** or **Medium** with **low confidence** (ambiguous or incomplete text) → flagged for human review rather than auto-scored — never force a confident-looking number out of an under-specified report.
- Reports scoring **Low** → logged normally, but still searchable/available if a later pattern emerges that retroactively makes them relevant.

This design deliberately biases toward **escalating uncertainty to a human** rather than resolving it automatically — consistent with the recall-over-precision principle formalized with real academic backing in Chapter 31.

Chapter 27 — Finding Recurring Precursor Patterns

RiskRadar's second major deliverable (beyond per-report classification) is discovering patterns *across* many reports.

Techniques

- **Clustering:** grouping reports with similar embeddings together, without needing pre-defined categories — good for surfacing an emerging pattern you didn't know to look for.
- **Embeddings-based similarity search:** given a new report, instantly find the most similar past reports ("this is the Nth similar report in the last N days") — the mechanism behind the "similar precursor" feature demonstrated throughout this handbook's worked examples.
- **Anomaly detection:** flagging when the *rate* of a particular hazard/activity/barrier-failure combination at a site suddenly increases relative to its own historical baseline — the mechanism behind "emerging pattern" alerts.
- **Trend analysis over time:** simple time-windowed counting (e.g., 30/60/90-day rolling windows) of precursor combinations, to distinguish a genuine emerging trend from ordinary random variation.
- **Association-rule mining:** discovering which combinations of factors (activity + location + shift + contractor status) co-occur more often than chance would predict.
- **Graph analysis:** representing hazards, activities, barriers, and locations as connected nodes, so relationships (not just frequencies) become visible — this is the technical foundation for the "SIF Precursor Chain" feature detailed fully in Chapter 38.

A simple worked example

```
Maintenance + Energy Isolation + Isolation-Verification-Failure
      appears in:
         Site A (3 times, last 45 days)
         Site B (2 times, last 30 days)
         Site C (1 time, last 10 days)
      ↓
Emerging Pattern flagged:
"Energy-isolation verification failures during maintenance have
appeared at 3 sites in the last 45 days — a rate higher than this
combination's historical baseline."
```

> **Why This Matters to RiskRadar:** This is the difference between a system that answers "was this one report dangerous?" and a system that answers "is something systemic going wrong across the organization?" — the second question is what turns RiskRadar from a classifier into the "safety intelligence platform" positioning described in Chapter 38.

References for this file

- PMC (PubMed Central), "Automatic identification of incidents involving potential serious injuries and fatalities (PSIF)" — pmc.ncbi.nlm.nih.gov/articles/PMC10998882/ (verified real peer-reviewed source informing the hybrid-extraction and recall-priority design choices in this chapter; full citation in Chapter 34)
- General NLP/LLM concepts in this chapter reflect standard, well-established practice in the field as of early 2026 and are not tied to a single proprietary source.

PART 06

Part 17–19: Data, Evaluation, and AI Safety

Chapter 28 — OIL Data Availability

What is publicly available

- OIL's **published safety statistics** at an aggregate level — for example, its FY2024–25 LTIFR of 0.071 (Chapter 5), training-coverage percentages, and general descriptions of its HSE Management System, reporting portals, and risk-assessment practices (HAZOP, QRA, JSA).
- OIL's **general operational and organizational profile** (Chapter 2) — business segments, locations, subsidiary structure.
- **Industry-wide, non-OIL-specific** datasets and research on safety-report NLP — real, usable, and public (Chapter 34).
- IOGP's **published Life-Saving Rules content, statistics, and guidance** (Chapter 15) — public and free to reference (though the icons and exact wording are trademark-protected and not to be altered — see IOGP's own usage terms).

What is likely internal / restricted

- **The actual free-text content of OIL's UA/UC/near-miss/incident reports.** This is the exact dataset the PS describes RiskRadar as needing to ingest — and it was **not found in any public source** during this handbook's research. This should be treated as OIL's internal, likely confidential operational data.
- **Exact incident volumes, exact SIF-labeled historical data, and exact site-level breakdowns.** Not publicly available.
- **The internal specifics of initiatives like "KAVACH"** beyond the general description that it aims to improve safety practices (Chapter 5).

The honest way to say this to a judge

> *"Our production target is OIL's actual historical UA/UC, near-miss, and incident report text. That dataset is not publicly available to us, so our prototype is built and demonstrated on a domain-calibrated synthetic dataset, modeled on OIL's confirmed reporting categories, IOGP's Life-Saving Rule taxonomy, and the published SIF research this handbook cites. The architecture is designed so OIL's actual authenticated report data could be substituted in directly, without redesigning the pipeline."*

This is not a weakness to hide — every hackathon team facing this exact PS has the exact same data gap. What differentiates a strong team is having a clear, honest, technically sound plan for it, which the rest of this Part provides.

Chapter 29 — Dataset Design

Proposed fields (aligned to the domain model in Chapter 21)

Field	Type	Example
`report_id`	string	`NS-2026-000123`
`report_type`	enum	Near Miss / UA / UC / Incident
`date`	date	`2026-03-14`
`location`	string (from a controlled site list, not free text)	`Field Site 4`
`activity`	enum (from Ch.14 taxonomy)	Maintenance
`narrative_text`	free text	*(the actual written report)*
`hazard`	enum	Stored/pressurized energy
`energy_type`	enum	Hydrocarbon pressure
`exposure_present`	boolean	true
`barrier`	enum	Positive isolation
`barrier_failure_type`	enum (Ch.11)	Unverified
`sif_potential_label`	enum	High / Medium / Low
`life_saving_rule`	array of enum (Ch.15)	[Energy Isolation]
`process_safety_relevant`	boolean	true
`actual_severity`	enum	None / First Aid / Medical Treatment / Lost Time / Fatality
`contractor_involved`	boolean	true

Which fields are categorical vs. free text

Everything except `narrative_text` (and a possible `notes`/free-text corrective-action field) should be a **controlled, enumerated value**, not free text — this is what makes the dataset usable for both training a classifier and building a reliable dashboard filter/aggregation layer. The free-text narrative is the *input*; the rest of the schema is the *output* your model is being trained (or, for the prototype, hand-labeled) to predict.

Chapter 30 — Synthetic Data Strategy

How to build a credible synthetic dataset

1. **Start from the Chapter 14 hazard taxonomy and Chapter 9's precursor examples** — every synthetic report should be traceable to a specific, named hazard/barrier/activity combination, not generated freely.
2. **Write narratives the way real field reports actually read** — specific, slightly clipped, technical, often missing punctuation polish. Compare the illustrative example in Chapter 20 as a style reference: *"Positive isolation was not verified with a pressure test before flange breaking commenced"* reads like a real HSE report; *"the worker was being unsafe"* does not.
3. **Deliberately build in the hard cases**, not just the obvious ones:
   - **Hidden SIF:** no injury, high potential (the isolation-verification example throughout this handbook).
   - **Non-SIF despite injury:** an actual minor injury (e.g., a soft-tissue strain from manual lifting) where the DEKRA-verified evidence (Chapter 8) says the potential for escalation is genuinely low.
   - **Ambiguous / incomplete:** a report missing key details needed to confidently score it — a good test of whether your confidence-scoring logic (Chapter 26) correctly flags uncertainty instead of guessing.
   - **Multi-hazard:** a single report touching two or three Life-Saving Rules at once (like the SIMOPS example in Chapter 9).
   - **Contradictory:** a narrative that says isolation was "completed" in one sentence but "not verified" in a later one — testing whether extraction correctly weighs the more specific, more recent, more operative statement.
4. **Aim for a realistic class imbalance**, not an artificially even dataset — see Chapter 31 for why this matters, and DEKRA's verified ~25%-of-recordables figure (Chapter 8) as a reasonable calibration anchor for what fraction of reports might plausibly carry real SIF potential.

Prototype validation vs. real-world validation — do not blur this line

	Prototype validation	Real-world validation
Data	Synthetic, hand-designed	OIL's actual historical, authenticated reports
What it proves	The pipeline runs correctly end to end; the logic behaves sensibly on designed test cases, including the hard cases above	Whether the system actually performs well on the messiness of real operational language and real base rates
What it does NOT prove	Real-world accuracy, real-world false-positive/false-negative rates	—

Never claim prototype-stage metrics as if they were production accuracy. If a judge asks "how accurate is your model," the correct answer distinguishes these two explicitly (see the full scripted answer in Chapter 41).

Chapter 31 — Measuring AI Performance

Why raw accuracy is dangerous for a rare-event problem like this

Suppose, out of 100,000 reports, **1,000 (1%) are truly SIF-potential** and 99,000 are not — roughly consistent with DEKRA's finding that SIF potential concentrates in a minority of events (Chapter 8). A model that predicts **"non-SIF" for every single report** would be **99% accurate** — and completely useless, because it would miss every single genuinely dangerous report. This is exactly why accuracy alone must never be the headline metric for this kind of problem.

The metrics that actually matter

- **Precision:** of the reports the model flagged as SIF-potential, what fraction genuinely were? (Precision = True Positives / (True Positives + False Positives)) — measures how much you can trust a positive flag.
- **Recall (Sensitivity):** of the reports that were genuinely SIF-potential, what fraction did the model actually catch? (Recall = True Positives / (True Positives + False Negatives)) — measures how much the model *misses*.
- **F1 score:** the harmonic mean of precision and recall — a single number balancing both, weighted equally.
- **F-beta / F2 score:** a variant of F1 that **weights recall more heavily than precision** — appropriate whenever missing a true positive is worse than a false alarm.
- **Confusion matrix:** a simple 2×2 (or larger) table of predicted vs. actual outcomes — the foundation all the above metrics are computed from.
- **PR-AUC (Precision-Recall Area Under Curve):** summarizes precision/recall trade-off performance across all possible decision thresholds — generally the more informative summary metric for rare-event/imbalanced classification problems like this one, compared to plain accuracy or even ROC-AUC.
- **ROC-AUC:** a related but distinct summary metric (true-positive rate vs. false-positive rate across thresholds) — useful, but can look artificially strong on heavily imbalanced data, so PR-AUC is usually reported alongside it for problems like this.

Why SIF detection should prioritize recall — with real, published backing

This is not just an intuition — it has been directly validated in real, peer-reviewed research on this **exact problem**. A 2024 peer-reviewed study (PMC10998882, full citation in Chapter 34) on automatic identification of PSIF (Potential Serious Injury and Fatality) from incident reports explicitly states that its safety subject-matter experts guided the team to **"prioritize the maximization of recall over precision, through the use of the F2 metric for hyperparameter tuning."** In plain terms: in a safety-critical setting, **missing a genuinely dangerous report (a false negative) is worse than sending an extra, ultimately-not-dangerous report for human review (a false positive)** — so it's correct, and independently validated by real published research on this exact task, to deliberately tune RiskRadar's threshold toward higher recall, even at some cost to precision.

> **Judge Tip:** This is one of the strongest, most specific things you can say in a Q&A: *"We're not guessing at this — a 2024 peer-reviewed study on this exact PSIF-classification problem found the same thing our safety logic assumes: recall should be prioritized over precision, using an F2 metric, because a missed high-potential report is a worse outcome than an extra report sent for human review."*

Chapter 32 — What If RiskRadar Is Wrong? (Failure Modes and Safeguards)

Failure modes

- **False negative:** a genuinely dangerous report is scored as low SIF potential and never escalated — the most serious possible failure mode for this system.
- **False positive:** a genuinely low-risk report is scored as high SIF potential — costs reviewer time, but is far less dangerous than a false negative.
- **Hallucination:** the AI extraction layer invents a hazard, barrier, or detail that the source text doesn't actually support (Chapter 24).
- **Bias:** the model performs worse on activities, sites, or report-writing styles under-represented in its training/synthetic data.
- **Overconfidence:** the model reports high confidence on a case it actually has weak evidence for.
- **Distribution shift:** the model was tuned on synthetic data (or one time period/site) and performs differently when it meets OIL's real, messier operational language.
- **Automation bias:** the human reviewer starts trusting the AI's score by default, and stops applying independent judgment — a well-documented risk in human-AI decision systems generally, and a serious one in a safety-critical context specifically.
- **Data leakage:** confidential OIL safety-report content being sent to, and potentially retained or logged by, an external third-party AI service without appropriate controls — a real deployment concern (see Chapter 36).

Safeguards

- **Confidence thresholds** that route low-confidence outputs to mandatory human review rather than an automated final answer (Chapter 26).
- **Evidence display** — every SIF score shown alongside the specific extracted text/fields that produced it (Chapter 25), so a reviewer can quickly sanity-check rather than blindly accept or reject.
- **Mandatory human review** for the highest-consequence categories, regardless of model confidence — never let the system "auto-clear" a report as definitely safe.
- **Escalation pathways** clearly defined for uncertain or borderline cases.
- **Audit logs** — a record of every score, the evidence behind it, and any human override, both for accountability and as a feedback source for improving the model over time.
- **Model monitoring** — tracking how the model's score distribution and confidence behave over time, to catch drift early rather than after it's caused a real missed case.
- **Feedback loops** — when a human reviewer overrides a score, that correction should be captured and eventually used to improve the system, not discarded.

> **NorthStar Must Know:** The single sentence that answers almost every "what if it's wrong" question a judge can ask: **"RiskRadar is a prioritization and decision-support system, not an autonomous safety authority — it is built to prefer escalating uncertainty to a human over quietly reassuring anyone that a report is safe."**

References for this file

- DEKRA's ~25%-of-recordables SIF-potential figure (Chapter 8) is used here as a realistic base-rate anchor for synthetic dataset design; source as cited in Chapter 3's References.
- PMC (PubMed Central), "Automatic identification of incidents involving potential serious injuries and fatalities (PSIF)" — pmc.ncbi.nlm.nih.gov/articles/PMC10998882/ — the direct source for the F2/recall-prioritization finding in Chapter 31. Full citation with authors and venue in Chapter 34.
- General precision/recall/F1/PR-AUC/ROC-AUC definitions in this chapter reflect standard, well-established machine-learning practice and are not tied to a single proprietary source.

PART 07

Part 20–23: Competitors, Research, Regulations, and Deployment

Chapter 33 — Existing Products (Verified Competitor Analysis)

Read this chapter carefully before your pitch. The single biggest risk to NorthStar's credibility is not the data gap covered in Chapter 28 — it's the fact that **at least one real, shipping commercial product already does something extremely close to RiskRadar's core function.** Knowing this, and having a sharp, honest answer for it, is more valuable than anything else in this handbook.

VelocityEHS — AI PSIF Insights (launched July 2025)

What it actually does, verified from VelocityEHS's own press materials: VelocityEHS's **"AI PSIF Insights"** analyzes existing incident and near-miss reports (no new forms, no workflow changes required) to identify the potential for temporary/permanent disabilities, amputations, fatalities, and other severe outcomes. It **automatically flags high-potential reports and sends notifications**, evaluates the completeness/clarity of incident descriptions and suggests improvements, and is marketed as **reducing manual triage time by 30–50%**. Its target market, per VelocityEHS's own materials, is mid-sized to enterprise organizations in "manufacturing, food & beverage, pharmaceuticals, and chemicals" — **oil & gas is not named as a primary target vertical.**

This is, functionally, a working commercial version of RiskRadar's Chapter 1/26 core capability, already in market as of mid-2025.

RiskRadar's honest differentiation against it: - VelocityEHS's own marketing does not describe IOGP Life-Saving Rule mapping, oil-and-gas-specific hazard taxonomy, or a barrier-failure/precursor-graph feature — RiskRadar's Chapter 15 rule-mapping and Chapter 38 precursor-chain features are not things VelocityEHS is shown to offer. - VelocityEHS is a general industrial-safety platform, not built around a specific national oil company's operating context (OIL's Northeast India onshore fields, its specific HSE Transformation Plan, its specific reporting-portal ecosystem).

Cority — Cortex AI

What it does, verified from Cority's own materials: Cority's **Cortex AI** includes 12+ AI "agents" across 30+ EHS use cases, explicitly including **"SIF scoring"** and **"predictive risk intelligence,"** alongside voice-to-text incident capture, image/document analysis for incident records, and permit-text analysis.

RiskRadar's honest differentiation: Same shape of argument as VelocityEHS — Cority's SIF scoring is general-purpose within its broad EHS+ platform, not shown to include IOGP Life-Saving Rule mapping or oil-and-gas-specific precursor-graph analytics.

Intelex — Input AI / Insight AI

What it does, verified from Intelex's own materials: AI-powered **incident pattern analysis** to identify recurring safety issues and leading-indicator trends, **predictive risk scoring**, an AI writing assistant for incident descriptions, and auto-fill for incident report fields. Independent third-party review sources describe Intelex's AI as requiring EHS-professional validation of recommendations rather than acting autonomously — consistent with the human-in-the-loop principle this handbook recommends throughout (Chapter 32).

DEKRA, EEI, Campbell Institute, IOGP

These are **not software competitors** — they are the **methodology sources** this handbook draws from directly (full treatment in Chapters 9–10, 15). DEKRA does offer paid consulting services and its own "SIF Potential Indicator" tool; it is a competitor in the sense of "an organization that already sells SIF-related risk assessment," but it is not an AI/NLP text-classification product built for automatically processing large volumes of free-text reports the way the PS specifies.

Comparison table

Product	Category	Confirmed AI SIF/PSIF capability	IOGP Life-Saving Rule mapping shown?	Oil & gas / OIL-specific?	RiskRadar's opportunity
VelocityEHS (AI	EHS software, AI	Yes — explicit PSIF	Not shown in public	Not a named target	Sharper oil-and-gas

PSIF Insights)	add-on	classification from report text	materials	vertical	taxonomy, LSR mapping, precursor graph, OIL-specific tuning
Cority (Cortex AI)	EHS+ platform, AI suite	Yes — explicit “SIF scoring”	Not shown in public materials	Not shown as oil-and-gas-specific	Same as above
Intelex (Input/Insight AI)	EHS platform, AI suite	Predictive risk scoring, pattern analysis (not SIF-labeled specifically in public materials)	Not shown	Serves “energy companies” among many verticals per third-party review	Focus, explainability, and precursor-chain depth
DEKRA SIF Prevention™ / SIF Potential Indicator	Consulting + assessment tool	Yes — methodology and a public online indicator tool	No (general cross-industry methodology)	No	RiskRadar operationalizes the *definition*, at report-processing scale, inside OIL’s workflow
IOGP Life-Saving Rules	Industry framework, not software	N/A	N/A (this *is* the framework)	Yes, oil & gas specific	RiskRadar automates the mapping *to* this framework

The honest positioning statement

> *“AI-assisted PSIF/SIF classification from incident-report text is not a novel idea in the abstract — VelocityEHS, Cority, and Intelex all now ship AI features aimed at exactly this problem, and DEKRA has published a methodology-driven public tool for it. What none of these publicly show is oil-and-gas-specific hazard and Life-Saving Rule mapping, barrier-failure extraction tuned to oil & gas operations, or a precursor-graph view built for a national oil company’s actual site/activity structure. RiskRadar’s contribution is not inventing AI-assisted SIF classification — it’s building the sharply-scoped, IOGP-aligned, explainable version of it, purpose-built for OIL’s workflow, as the PS specifically asks for.”*

> **Judge Tip:** Say this proactively, before a judge brings it up. A team that says “nobody else has done this” and gets corrected by a judge who happens to know about VelocityEHS’s July 2025 launch loses far more credibility than a team that opens with “here’s how we’re different from what already exists.”

Chapter 34 — Academic Research (Verified, With Real Citations)

Directly on-topic: automatic PSIF/SIF classification from incident-report text

"Automatic identification of incidents involving potential serious injuries and fatalities (PSIF)" — published via PubMed Central (PMC10998882). This paper explores ML-powered automatic identification of PSIF from incident reports to assist workplace-safety risk identification. **Method:** encodes heterogeneous incident-record fields through multiple techniques, including generating dense embeddings from free-text fields via a **BERT-based transformer model**, combined with regular-expression and term-frequency encoding for quantity-based fields, then applies **XGBoost** for binary classification. **Key finding directly relevant to RiskRadar's design (Chapter 31):** the team's safety subject-matter experts specifically guided prioritizing **recall over precision**, tuned via the **F2 metric**. **Relevance:** this is, functionally, an almost direct academic analog of what RiskRadar needs to build — read this paper closely if your AI/ML members do nothing else from this reading list.

Directly relevant: transformer-based classification of occupational fatality/accident narratives

"Multimodal and Explainable Deep Learning for Occupational Accident Classification Using Transformer-LSTM Architectures" (2026, published in *Buildings*, DOI: 10.3390/buildings16091642). **Method:** a Hybrid Transformer-LSTM framework classifying occupational fatalities using **14,914 records from the U.S. OSHA database**, combining BERT-based embeddings (semantic extraction from narrative text) with Bidirectional LSTM layers encoding temporal and spatial features. **Notable design choice, directly relevant to RiskRadar's explainability requirement (Chapter 25):** the study is explicitly grounded in the **Swiss Cheese Model** (Chapter 12), treating different data modalities as proxies for different layers of system risk. It also specifically tests whether the model relies on keyword shortcuts rather than genuine semantic understanding — finding it generalizes to narratives where the "obvious" keyword is absent (e.g., correctly identifying "loss of balance from a high platform" as fall-related without the word "fall" present). **Result:** 84.56% accuracy, a 5.33-percentage-point improvement over a unimodal BERT baseline. **Relevance:** strong evidence that combining textual semantic extraction with structured contextual features (time, location) outperforms text-only classification — directly informs RiskRadar's domain model (Chapter 21), which deliberately includes location and activity as structured fields alongside the narrative.

Foundational: injury-narrative auto-coding

NIOSH (the U.S. National Institute for Occupational Safety and Health) ran a public **2018 injury-narrative auto-coding research initiative**, motivated by the observation that manually coding the large volume of free-text injury narratives collected in medical records and surveillance systems is expensive, slow, and error-prone. Subsequent published work (e.g., BERT-based approaches built on Hugging Face and cloud ML infrastructure) has continued this line of research, applying transformer models to auto-code injury narratives. **Relevance:** this is the earliest, most institutionally authoritative precedent for "government/industry safety body recognizes free-text safety narratives as a real NLP opportunity, not just a hackathon idea" — useful context for a judge who asks "is this a real research area or something you invented for the pitch."

Broader academic landscape (from a verified systematic review)

A published **systematic review** (PMC9521307, PRISMA-based, reviewing 27 of 210 identified studies) on text-mining and NLP applications for occupational-injury narrative extraction confirms this is now an established, if still-developing, research area, spanning multiple techniques: Bayesian classification mapping narratives to OSHA codes; random-forest classification of MSHA (Mine Safety and Health Administration) narratives; BERT-based approaches specifically noted as improving on earlier methods; Latent Semantic Analysis (LSA) for finding similar reports; Latent Dirichlet Allocation (LDA) topic modeling applied to predicting rail-incident severity; and BERT-based classification of near-miss events specifically in **construction-site reports** (a close industry analog to oil & gas field operations).

A notable, India-relevant gap in the literature

One 2026 study, "Text Analysis of Workplace Accident Chronology for Hazard Identification Using BERT and Random Forest Algorithms," explicitly notes that while NLP/ML has gained real traction for safety-narrative analysis in "other high-risk industries," **its application to mining accident data "within the Indian context, remains limited." This is a genuinely useful, honest data point for your pitch**: it shows independent, published academic acknowledgment that India-specific, high-risk-industry safety-NLP work is an under-served research area — which strengthens (without overclaiming) the case that a well-executed OIL-specific system has real novelty value, even though the general technique (AI/NLP for safety-report classification) is not itself new.

> **NorthStar Must Know:** Use this chapter's findings precisely. The correct claim is **"AI/NLP-based PSIF/SIF classification is an active, validated academic and commercial field — RiskRadar's specific contribution is the IOGP-aligned, OIL-specific, explainable version of it."** The incorrect claim is **"nobody has ever done AI safety-report classification before."** A judge who has read even one of the papers above will catch the second claim immediately.

Chapter 35 — Regulations and Standards (Verified)

Law vs. standard vs. guidance vs. company policy — get this distinction right

- **Law:** enacted by a legislature, legally binding on everyone within its scope (e.g., India's labour codes).
- **Standard:** a detailed technical specification, often issued by a government-linked technical body; may or may not be legally mandatory depending on whether it has been formally adopted into a licensing/regulatory framework.
- **Guidance / industry framework:** authoritative but not, by itself, legally binding — e.g., IOGP's Life-Saving Rules, which IOGP itself describes as supporting, not replacing, a company's actual management system and any applicable law.
- **Company policy:** OIL's own internal rules and procedures, which may (and generally should) meet or exceed the applicable legal/standard floor.

India's oil & gas safety regulatory landscape (verified)

- **OISD — Oil Industry Safety Directorate.** Established **1986** under the **Ministry of Petroleum and Natural Gas (MoPNG)**. A technical directorate that develops and coordinates safety **standards** for the design, layout, and operation of oil & gas facilities in India, and monitors industry safety performance. OISD publishes **over 200 standards**, identified by number (e.g., **OISD-STD-105** for work-permit systems; **OISD-STD-108** for oil storage and handling; **OISD-117** for fire safety at retail fuel outlets; **OISD-118** for facility layout and inter-equipment safe distances). OISD itself is described (by sources reviewing its role) as a **regulatory-but-not-statutory body except in parts of the upstream petroleum sector** — but critically, **around 11 of its standards have been formally adopted into PESO's rules**, which makes compliance with those specific standards legally mandatory to that extent. OISD assists India's **Safety Council**, chaired by the Secretary, Ministry of Petroleum & Natural Gas, with representation from PSU chief executives, the Chief Controller of Explosives, the Director General of Mines Safety, and others, meeting at least annually to review industry safety performance.
- **PESO — Petroleum and Explosives Safety Organisation.** Administers licensing and enforcement under India's **Petroleum Act, 1934** and the **Explosives Act**, covering petroleum storage, explosives, compressed gases, pressure vessels, and hazardous premises. PESO issues and renews the licenses that let a petroleum facility legally operate, and has the authority to inspect and to suspend or cancel a license for non-compliance.
- **PNGRB — Petroleum and Natural Gas Regulatory Board.** Downstream-focused technical regulator, with standards covering pipelines (both natural gas and petroleum-product pipelines), city gas distribution networks, LNG terminals, and petroleum installations (its "T4S" technical standards have been periodically updated).
- **DGMS — Directorate General of Mines Safety.** Relevant specifically to drilling and mining-adjacent operations.
- **State Factory Inspectorates.** Enforce factory-level occupational safety law at the state level.

The (currently still-consolidating) national labour law framework

India has been consolidating numerous pre-existing labour laws (including the historic Factories Act, 1948) into four new labour codes, one of which — the **Occupational Safety, Health and Working Conditions Code, 2020** — is specifically aimed at workplace safety and health. **Public information on the code's exact rollout/enforcement status could not be independently confirmed as fully in force for this handbook** — implementation of India's labour codes generally has depended on individual states notifying their own supporting rules, and this has been a gradual, multi-year process. If this matters for your pitch, verify the current in-force status directly (e.g., via the Ministry of Labour & Employment) close to your presentation date, rather than stating it as settled here.

International standards worth knowing (context, not something to claim RiskRadar "certifies")

- **ISO 45001** — the international standard for Occupational Health and Safety Management Systems.
- **ISO 14001** — the international standard for Environmental Management Systems.
- **ISO 31000** — the international standard providing general risk-management principles and guidelines.
- **OSHA** — the U.S. Occupational Safety and Health Administration; referenced repeatedly in this handbook because much of the available public safety-narrative research data (Chapter 34) is OSHA-derived, and because IOGP itself cross-checked its Life-Saving Rules content against OSHA data (Chapter 15).

NorthStar Must Know: RiskRadar does not certify OIL's compliance with any of the above, and should never be pitched as a compliance/certification tool. It is a decision-support and prioritization layer that can *help* HSE teams act on the kind of risk information these regulatory bodies care about — a materially different, and more honest, claim.

Chapter 36 — How OIL Could Actually Deploy RiskRadar

Deployment options

- **On-premise:** RiskRadar runs entirely within OIL's own infrastructure — strongest data-control story, most deployment effort.
- **Private cloud:** RiskRadar runs in a cloud environment dedicated to OIL (or contracted with strong isolation guarantees) — a common middle ground for safety-sensitive public-sector enterprises.
- **Enterprise cloud / SaaS:** RiskRadar runs as a shared, multi-tenant service — least deployment effort, but the weakest data-control story for confidential safety-report content, and the option most likely to draw a hard "would OIL actually send this data there?" question from a judge.

Integration points

- **API integration** with OIL's existing HSSE reporting platform, so RiskRadar can process new reports as they're submitted, rather than requiring a separate manual upload step.
- **SSO (Single Sign-On)** so HSE staff use their existing OIL credentials.
- **RBAC (Role-Based Access Control)** so, for example, a site-level safety officer sees their site's queue while a corporate HSE leader sees the enterprise-wide view.
- **Audit logs** — a full record of what was scored, what evidence was shown, and what a human reviewer did with it (also a Chapter 32 AI-safety requirement, not just a deployment detail).

Why confidential safety-report content needs special handling

Safety-report narratives can contain sensitive operational detail (specific equipment vulnerabilities, specific personnel involved, specific site security-relevant layout information). A judge's likely question — **"would OIL actually send confidential incident reports to a public LLM API?"** — deserves a direct, technically grounded answer, not a dodge:

> *"For a production deployment, RiskRadar's extraction layer would run on infrastructure OIL controls or contracts under strong data-processing terms — private cloud or on-premise, not a consumer-facing public LLM endpoint with no data agreement. For the hackathon prototype, we're transparent that we may use an off-the-shelf LLM API for demonstration purposes only, using entirely synthetic data — never real OIL report content — specifically because that data-handling question doesn't have a good answer yet at prototype stage."*

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PART 08

Part 24–25: Product Design and the Killer Feature

Chapter 37 — RiskRadar's Dashboard: Screen by Screen

The design principle for every screen: **answer "what needs my attention?" before answering "what does the data look like?"** A dashboard full of pie charts with no clear point of view is a weak demo, however polished — see the red-team analysis in Chapter 42 for exactly this failure mode.

#	Screen	User	Purpose	What it shows	Decision it supports
1	**Executive Overview**	Corporate HSE leadership	"How is SIF-precursor risk trending across the whole organization?"	Enterprise-wide SIF-potential trend, top 3 sites/activities by precursor density, headline emerging-pattern alerts	Where to direct enterprise-level attention or resourcing
2	**HSE Priority Queue**	Site/corporate HSE reviewer	"Which reports need my attention today?"	A ranked list — not a flat table — of reports by SIF-potential score and confidence, newest high-priority first	What to review right now
3	**Report Detail**	HSE reviewer	"Why was this report flagged?"	Full extracted evidence: hazard, activity, exposure, barrier, barrier failure, confidence, source text highlighted	Confirm, override, or escalate the automated score
4	**SIF Explanation**	HSE reviewer, judges	"Show your work."	The step-by-step chain from Chapter 20/38, evidence-linked	Trust and auditability
5	**Life-Saving Rule Mapping**	HSE reviewer, site manager	"Which of the 9 rules is this connected to?"	Rule name, confidence, and the specific text that triggered the mapping	Deciding whether this needs Life-Saving-Rule-specific intervention (e.g. a targeted toolbox talk)
6	**Precursor Patterns**	HSE reviewer, investigators	"Has this happened before?"	Similar past reports (via embeddings similarity, Chapter 27), with dates and locations	Whether this is an isolated case or a recurring pattern deserving investigation
7	**Site Comparison**	Site/corporate HSE leadership	"Which sites carry the most concentrated risk?"	Sites ranked by precursor *density*, not raw count (Chapter 9)	Where to prioritize site visits, audits, or resourcing
8	**Activity Analysis**	HSE leadership	"Which activities carry the most SIF potential?"	Activities (maintenance, drilling, lifting, etc.) ranked by precursor density	Where to prioritize procedure review or targeted training
9	**Barrier Failures**	HSE leadership, process safety	"What controls are actually failing in practice?"	Barrier-failure-type breakdown (missing/failed/unverified/bypassed — Chapter 11), by hazard and site	Which specific control systems need reinforcement
10	**Trends / Emerging Patterns**	HSE leadership	"Is something new starting to happen?"	Time-windowed pattern detection output (Chapter 27)	Whether a fresh intervention is needed before a real SIF occurs
11	**Investigation**	Investigators	"Give me a fast	Pre-populated evidence extract for	Accelerating, not

	View		starting point."	a flagged report, structured for handoff into a formal RCA process (Chapter 19)	replacing, the human investigation

Common Mistake: Building screens 7–10 (all the aggregate/analytics views) before screens 2–4 (the per-report priority queue and explanation) are solid. The per-report experience is what a judge will actually click through during your demo — get that right first.

Chapter 38 — The Killer Feature: The SIF Precursor Chain

This is the single feature most worth polishing for your demo, because it is the clearest, most concrete demonstration that RiskRadar is reasoning about safety, not just running a classifier.

The concept

Instead of showing a bare score:

```
🔴 SIF Potential: 91%
```

show the full reasoning chain behind it:

```
Activity:               Hydrocarbon line maintenance
        ↓
Hazard:                  Stored/residual pressure
        ↓
Barrier:                 Positive isolation
        ↓
Barrier Failure:         Verification not performed
        ↓
Exposure:                Worker positioned at flange during break
        ↓
Potential Consequence:   Uncontrolled release — chemical exposure,
                          fire, or line-of-fire injury, up to fatal
        ↓
Life-Saving Rule:        Energy Isolation
        ↓
Pattern:                 3rd similar precursor in the last 45 days,
                          across 2 sites
        ↓
Recommended Action:      Escalate for review; consider a targeted
                          isolation-verification toolbox talk at the
                          affected sites
```

Why this is stronger than a bare classification score

- **It is auditable.** A reviewer can check each link, not just accept or reject a single number.
- **It is teachable.** The chain itself doubles as a training artifact — "here's exactly what an isolation-verification failure looks like in report language" is directly useful for a toolbox talk.
- **It survives the hardest judge question.** "Why did your model flag this?" has a specific, evidence-linked answer at every step, rather than "the model said so."
- **It naturally extends into the precursor graph (Chapter 27).** Each chain is one path through a much larger graph connecting hazards, barriers, activities, and locations — so the same underlying data structure that explains one report also powers the site/activity-level pattern discovery in screens 6–10 of Chapter 37.

How this reframes RiskRadar for your pitch

Rather than: *"Our AI classifies safety reports."*

Say: *"RiskRadar has four layers — it Understands what a report describes, Assesses whether it carries SIF potential, Explains exactly which hazard, exposure, and barrier failure caused that assessment, and Discovers whether the same precursor is repeating elsewhere. That's what turns a classifier into a safety-intelligence platform."*

```
🧠 Understand → 🚨 Assess → 🛡 Explain → 🔎 Discover → ⛑ HSE Action
```

> **Judge Tip:** If you can only build one thing to a truly polished, demo-ready standard given limited hackathon time, build the SIF Precursor Chain for a handful of well-chosen synthetic example reports — not the broadest possible feature set. This single artifact does more work in a live demo than any amount of extra dashboard chart types.

PART 09

Part 26–29: Team, Execution, and Judge Preparation

Chapter 39 — What Each NorthStar Member Should Learn

AI/ML member

Must know: Chapters 8–10 (SIF fundamentals and methodologies), 21 (domain model), 22–27 (NLP/LLM/hybrid architecture, scoring, precursor detection), 31–32 (evaluation, failure modes), 34 (research papers — read PMC10998882 closely). **Should know:** Chapters 14–15 (hazard taxonomy, Life-Saving Rules) well enough to design the label schema. **Can skip:** Deep detail on OIL's financial/corporate history (Chapter 2's early sections) and the incident-investigation methodology names in Chapter 19 beyond a passing familiarity.

Backend member

Must know: Chapter 3 (Ch.3 — data will describe real operational stages), Chapter 6 (HSE workflow — this is what your API needs to model), Chapter 21 (domain model — this is your schema), Chapters 28–29 (data design), Chapter 36 (deployment/integration). **Should know:** Chapter 25 (hybrid architecture — you're building the pipeline that connects the layers). **Can skip:** Deep IOGP rule-by-rule detail (Chapter 15) beyond knowing the 9 rule names exist as an enum.

Frontend member

Must know: Chapters 6–7 (HSE workflow and vocabulary — you're building the UI HSE staff will actually use), Chapter 21 (what fields exist to display), Chapter 37 (the 11 screens), Chapter 38 (the killer feature — this is your hero screen). **Should know:** Chapter 9's worked precursor examples, to build realistic-feeling mock data for your own testing. **Can skip:** The ML evaluation math in Chapter 31.

HSE / domain researcher

Must know: Everything in Parts 3–11 (Chapters 4–20) — you are the team's safety conscience, and your job in a Q&A is to catch the moment a teammate says something that would make a real HSE professional wince. **Should know:** Chapter 33 (competitors) well enough to explain how DEKRA/EEI/IOGP relate to each other without conflating them. **Can skip:** Deep NLP model internals (Chapter 23) beyond a conceptual level.

Product / UX member

Must know: Chapters 1 (the PS itself), 37–38 (dashboard and killer feature), 6 (HSE workflow — your screens must map onto real steps). **Should know:** Chapter 9's worked examples, to design realistic mock content. **Can skip:** Deep ML metric math (Chapter 31) beyond being able to explain "why recall matters" in plain English.

Presentation / business member

Must know: Chapter 1 (PS), the "Final Mental Model" section in File 10 (30-second and 2-minute explanations, memorized close to word-for-word), Chapter 33 (competitors — you will be asked this), Chapter 41 (Judge Q&A), Chapter 42 (red-team — know your own weaknesses cold). **Should know:** Chapter 2 (OIL) well enough to speak fluently about the company without notes. **Can skip:** Model architecture detail — defer confidently to your AI/ML teammate rather than guessing.

Chapter 40 — From Research to Prototype: A Build Order

```
1. Research (this handbook)
        ↓
2. Domain model (Chapter 21 — lock this early; everyone builds against it)
        ↓
3. Synthetic dataset (Chapters 29–30 — build the hard cases, not just the easy ones)
        ↓
4. Baseline classifier (a simple model first — don't start with the most
   complex possible architecture; get an end-to-end pipeline working, then improve it)
        ↓
5. AI extraction + hybrid scoring pipeline (Chapters 25–26)
        ↓
6. Dashboard (Chapter 37 — build screens 2–4 before 7–10, per the Common Mistake note there)
        ↓
7. Integration (connect real pipeline output to the real dashboard, not mocked data)
        ↓
8. Testing (deliberately run your hardest synthetic cases from Chapter 30 through the live system)
        ↓
9. Demo script (choose 1–2 reports and rehearse the SIF Precursor Chain walk-through, Chapter 38, until it's
smooth)
        ↓
10. Pitch (Chapter 1's "what success looks like," plus Chapter 41's Q&A prep)
```

What to build first, if time is short: a working, explainable SIF Precursor Chain (Chapter 38) for a small, well-chosen set of synthetic reports, connected end-to-end from raw text to dashboard display. A narrow, deep, honest demo beats a broad, shallow one every time in this kind of evaluation.

Chapter 41 — Judge Q&A

Each entry: a tough question, a strong answer specific to RiskRadar, a likely follow-up, and a strong follow-up answer.

Problem understanding

Q: In one sentence, what does RiskRadar actually do? A: *"It reads OIL's existing safety reports and finds the small number that quietly show fatal potential — even when nobody was hurt — then maps them to the relevant IOGP Life-Saving Rule and shows where the same precursor is recurring."* Follow-up: **"How is that different from a normal incident dashboard?"** A: *"A normal dashboard ranks by what happened. RiskRadar ranks by what could have happened — which, per DEKRA's published research, is a substantially different ranking."*

Q: What does "SIF potential" actually mean, precisely? A: *"DEKRA's own definition, which we adopt: a serious injury or fatality is reasonably likely to occur if the exposure continues uncontrolled. The test is: if this exact situation repeated dozens or hundreds of times, would you reasonably expect it to eventually produce a serious injury?"* Follow-up: **"Isn't that subjective?"** A: *"It requires judgment, yes — which is exactly why our scoring is built from concrete extracted signals (energy present, barrier status, exposure, proximity) rather than a single subjective label, and why low-confidence cases are routed to a human rather than auto-decided."*

OIL and domain knowledge

Q: Why OIL specifically? What do you actually know about their operations? A: *(Answer fluently from Chapter 2 — OIL's Maharatna status, its Northeast India core operations plus Rajasthan/offshore acreage and international blocks, NRL refining subsidiary, its confirmed HAZOP/QRA/JSA risk-assessment practice, its FY24–25 best-ever LTIFR of 0.071, and its active HSE Transformation Plan Phase-II — cite the Annual Report.)* Follow-up: **"How would this fit into what they're already doing?"** A: *"OIL is actively in Phase-II of its own HSE Transformation Plan, explicitly building a more robust HSE Management System right now — RiskRadar is a natural complement to that effort, not a competing initiative, and it plugs in after their existing near-miss reporting portals, not instead of them."*

Q: What's a UA/UC/near-miss, and why do you treat near-misses seriously? A: *(Use Chapter 7's definitions and Chapter 9's "zero injury can still be extremely dangerous" reasoning.)*

AI and technical approach

Q: Why do you need AI? Couldn't you just use keyword search? A: *(Use Chapter 22's "isolation completed" vs. "isolation was not verified" example directly — it is the single strongest, most concrete answer in this handbook for this exact question.)*

Q: What happens if the LLM hallucinates? A: *(Use Chapter 25's answer directly: the LLM only extracts evidence into a structured shape; a separate, auditable scoring layer operates on that structured evidence, so a hallucinated field that doesn't map to the taxonomy doesn't silently produce a wrong score.)*

Q: How accurate is your model? A: *"We distinguish two different claims. On our synthetic held-out test set, [state your actual measured numbers here once you have them] — but we're explicit that synthetic-data performance is a pipeline-functionality proof, not a production-accuracy claim. Real accuracy would require validation against OIL's actual historical, labeled reports, which we don't have access to."* Follow-up: **"So you can't actually prove it works?"** A: *"We can prove the pipeline correctly implements the extraction-to-score-to-explanation logic, including on deliberately hard cases — ambiguous reports, multi-hazard reports, contradictory narratives. What we can't yet prove is real-world accuracy at OIL's actual scale and language patterns, and we say so directly rather than overclaiming it."*

Q: Why not just use ChatGPT/an LLM for everything? A: *(Use Chapter 25's hybrid-architecture reasoning.)*

Data

Q: Where did your training data come from? A: *(Use Chapter 28's scripted honest answer directly, word for word if needed.)* Follow-up: **"Then how do you know your accuracy claims mean anything?"** A: *"We don't claim production accuracy from synthetic data — see the answer above. We built the synthetic set specifically to include hard cases (Chapter 30) precisely so pipeline-level testing is meaningful, even though it can't substitute for real-world validation."*

Q: What if OIL never gives you real data? A: *"Then RiskRadar remains a validated prototype architecture rather than a deployed production system — which is an honest, common outcome for a hackathon-stage project, and the architecture is specifically designed (Chapter 28) so real data could be substituted in later without a redesign."*

Safety and correctness

Q: What happens if your AI incorrectly labels a dangerous report as low risk? A: *(Use Chapter 32's failure-mode/safeguard framing — false negative is the most serious failure mode; safeguards include confidence thresholds, evidence display, and mandatory human review for high-consequence categories regardless of score.)*

Q: How will this actually prevent a fatality? A: *"We don't claim it predicts or prevents an individual fatality — that would be an unsupportable claim. Its mechanism is earlier visibility: identifying SIF-potential reports, explaining the specific precursor and barrier failure, detecting recurrence, and prioritizing that pattern for HSE intervention — moving from retrospective counting toward proactive precursor management."*

Q: Are you replacing the HSE team? A: *"No — RiskRadar is a prioritization and decision-support system. Every high-consequence or low-confidence case is routed to a human, and the system is designed to escalate uncertainty rather than resolve it automatically. See Chapter 32."*

Competition and novelty

Q: How is this different from existing EHS platforms? A: *(Use Chapter 33's full comparison table and honest positioning statement — do not claim novelty of AI-based SIF classification itself; claim the oil-and-gas-specific, IOGP-aligned, explainable version of it.)*

Q: Isn't VelocityEHS already doing this? A: *(This question rewards you for having read Chapter 33 closely — answer confidently: "Yes, VelocityEHS launched an AI PSIF Insights feature in July 2025 that analyzes incident/near-miss text for SIF potential — we know about it and we think that's evidence this is a real, validated problem space, not evidence RiskRadar has nothing to add. What their public materials don't show is IOGP Life-Saving Rule mapping or an oil-and-gas-specific barrier-failure taxonomy, which is where RiskRadar is scoped.")*

Deployment and security

Q: Would OIL actually send confidential safety reports to your AI? A: *(Use Chapter 36's scripted answer — on-prem/private-cloud for production, synthetic-only data for the hackathon demo.)*

Q: How would OIL actually deploy this at scale, across all their sites? A: *(Use Chapter 36's deployment options and integration points — API integration with the existing HSSE platform, SSO, RBAC, audit logs.)*

Business value and scalability

Q: What's the ROI? How much does this save OIL? A: *"We haven't modeled a specific rupee figure, and we'd rather say that honestly than invent one. The mechanism for value is faster triage (comparable competitor products claim 30–50% reduction in manual triage time) and earlier intervention on the highest-potential precursors — which is where the real cost of a SIF event (human, operational, and financial, as covered in Part 3 of the original problem framing) would otherwise land."*

Q: Can this scale across all of OIL's operations — onshore, offshore, refining, pipelines? A: *"The architecture is designed to, because the underlying hazard taxonomy (Chapter 14) is built around recurring hazard categories — stored energy, line of fire, confined space — that show up across every stage of the lifecycle, not around one specific piece of equipment. The dataset and fine-tuning would need to reflect each operating context, but the pipeline design doesn't need to change."*

Ethics and human oversight

Q: What stops HSE staff from just trusting the AI score and stopping independent thinking? A: *(Use Chapter 32's "automation bias" failure mode directly, and the evidence-display / explanation-chain design in Chapters 25 and 38 as the concrete mitigation — showing evidence, not just a score, is specifically meant to keep a human engaged rather than passively accepting a number.)*

Q: Could this be used to blame or punish individual workers? A: *(Use Chapter 12's Swiss Cheese Model reasoning: "RiskRadar flags active-failure signals in report text, but we specifically teach our own team, using James Reason's Swiss Cheese Model, that an active failure is rarely the whole story — a well-designed downstream investigation process should always ask what latent organizational conditions made that failure more likely. RiskRadar is built to accelerate finding the precursor, not to assign individual blame.")*

Chapter 42 — Red-Team: Attack Your Own Idea

Be honest here — a judge who finds a weakness you hadn't already identified is far more damaging than one who raises a weakness you'd already named and mitigated.

Biggest weakness: data

You do not have access to OIL's real report data, and this is the first thing a sharp judge will probe (Chapter 28). **Mitigation:** the honest, scripted answer in Chapter 28, plus a genuinely well-designed synthetic dataset (Chapter 30) that shows you understand what real data *should* look like, even without having it.

AI weakness: an existing, shipping competitor already does the core function

VelocityEHS's AI PSIF Insights (Chapter 33) is a real, launched, marketed product doing something very close to RiskRadar's classification core. **Mitigation:** don't hide this — lead with it (Chapter 33's honest positioning statement), and make sure your actual differentiation (IOGP rule mapping, oil-and-gas taxonomy, precursor-chain explainability, OIL-specific tuning) is genuinely built, not just claimed.

Domain weakness: none of you are HSE professionals

Mitigation: this handbook exists specifically to close that gap — the depth of Parts 3–11 should let any team member speak credibly, and your dedicated HSE/domain researcher (Chapter 39) should be ready to field the hardest domain questions directly.

Deployment weakness: no real integration with OIL's actual HSSE platform

Mitigation: be upfront that this is a prototype architecture with a clear integration plan (Chapter 36), not a claim of an existing integration.

Competitor weakness: "why would OIL build this instead of buying VelocityEHS/Cority?"

Mitigation: the honest answer is that a purpose-built, IOGP-aligned, India-oil-and-gas-specific tool can be more precisely tuned to OIL's exact taxonomy and workflow than a horizontal platform serving "manufacturing, food & beverage, pharmaceuticals, and chemicals" (VelocityEHS's own stated target verticals) — and that a hackathon prototype is a starting point for exploring that build-vs-buy question, not a claim that RiskRadar is already superior to a mature commercial product.

Regulatory weakness: you have not mapped RiskRadar to specific OISD/PESO compliance requirements

Mitigation: be explicit that RiskRadar is a decision-support tool, not a compliance-certification tool (Chapter 35's closing note) — this actually reduces your regulatory exposure rather than increasing it, since you're not claiming to certify anything.

UX weakness: a beautiful dashboard with no real intelligence behind it

Mitigation: Chapter 37's explicit build-order note — build the per-report SIF Precursor Chain (screens 2–4, Chapter 38) before the aggregate analytics screens, and make sure your demo leads with reasoning, not chart variety.

Credibility weakness: overclaiming in the pitch

The single most damaging thing NorthStar could do is claim the system "prevents fatalities," "predicts" a specific future event, or reports an accuracy number without the synthetic-vs-real caveat. **Mitigation:** every "what NOT to claim" callout throughout this handbook exists specifically to prevent this — before your final pitch, do one pass of your slides and script looking specifically for these phrases.

> **Judge Tip:** The strongest possible thing to say when a judge raises a weakness you've already identified is simply: *"Yes — that's a real limitation, here's how we've thought about it..."* rather than getting defensive. Judges consistently rate honest, self-aware teams higher than teams that pretend not to have weaknesses.

PART 10

Part 30–32: Knowledge Check, Quick Reference, and the Final Mental Model

Chapter 43 — NorthStar Certification Test

Answer key follows each level. No peeking until you've genuinely tried.

Level 1 — Basic (20 questions)

1. What does SIH Problem Statement 26165 ask NorthStar to build, in one sentence?
2. What does "OIL" stand for in this project?
3. What are the four report types the PS names?
4. What does HSE stand for?
5. What does HSSE add to HSE?
6. What is the difference between an Unsafe Act and an Unsafe Condition?
7. What is a Near Miss?
8. What does SIF stand for?
9. What is the difference between actual severity and potential severity?
10. What does LTIFR measure?
11. What does PTW stand for?
12. What does LOTO stand for?
13. How many IOGP Life-Saving Rules are there today?
14. Name three of the nine Life-Saving Rules.
15. What is a "barrier" in safety terms?
16. What is the difference between process safety and personal safety?
17. What does JSA stand for?
18. What is OIL's confirmed FY2024–25 LTIFR figure?
19. Where is OIL's field/registered headquarters located?
20. What does "SIF potential" mean, in DEKRA's own definition?

Level 1 Answer Key

1. Build an AI/NLP system that reads OIL's free-text safety reports, classifies each for SIF potential, maps it to an IOGP Life-Saving Rule, and surfaces recurring precursor patterns on a dashboard.
2. Oil India Limited.
3. Unsafe Act (UA), Unsafe Condition (UC), Near Miss, Incident.
4. Health, Safety, and Environment.
5. Security.
6. A UA is a person's behavior; a UC is a state of the physical environment/equipment.
7. An event where something went wrong or almost went wrong, but no injury or damage occurred.
8. Serious Injury or Fatality.
9. Actual severity is what really happened; potential severity is what could have happened if circumstances were slightly worse.
10. Lost Time Injury Frequency Rate.
11. Permit to Work.
12. Lockout/Tagout.
13. Nine.
14. Any three of: Bypassing Safety Controls, Confined Space, Driving, Energy Isolation, Hot Work, Line of Fire, Safe Mechanical Lifting, Work Authorisation, Working at Height.
15. Anything that stands between a hazard and a person, preventing harm.
16. Personal safety protects individuals from direct, acute hazards; process safety prevents loss of containment events

that can affect many people.

17. Job Safety Analysis.
18. 0.071 (best-ever, per its FY2024–25 Annual Report).
19. Duliajan, Assam.
20. A serious injury or fatality is reasonably likely to occur if the exposure continues uncontrolled.

Level 2 — Intermediate (20 questions)

1. Why does DEKRA's research say fatality rates and minor-injury rates don't move together?
2. What fraction of OSHA recordables does DEKRA say carry realistic SIF exposure potential?
3. What is a "SIF precursor," in DEKRA's precise definition?
4. What is the "dice roll" test for judging SIF potential?
5. What is EEI, and what sector is its SIF work built for?
6. What is the EEI Safety Classification and Learning (SCL) Model built around?
7. Name the six ways a barrier can fail.
8. What is the Swiss Cheese Model, and who developed it?
9. What is the difference between an "active failure" and a "latent condition"?
10. What does a Bow-Tie diagram show, left to right?
11. How many Life-Saving Rules existed before the 2017–18 simplification, and how many after?
12. What percentage of 2008–2017 fatalities did excavation-related incidents represent, per IOGP — and why does that matter for how the Rules are structured today?
13. What height does IOGP recommend as a rule-of-thumb threshold for fall protection?
14. Why is "isolation completed" different from "isolation verified"?
15. What is RAG, in the context of LLMs, and why is it useful for Life-Saving Rule mapping?
16. Why is accuracy a dangerous headline metric for SIF classification?
17. What metric did a real 2024 peer-reviewed PSIF-classification study prioritize, and why?
18. Name three real commercial products with AI-powered SIF/PSIF-related features.
19. What is OISD, and when was it established?
20. What is the difference between OISD and PESO?

Level 2 Answer Key

1. Because the underlying causes of minor injuries and of fatalities are often different — reducing minor-injury rates doesn't proportionally reduce fatal risk.
2. About 25%.
3. A high-risk situation in which management controls are either absent, ineffective, or not complied with.
4. If this exact situation repeated dozens or hundreds of times, would it be reasonable to conclude a serious injury or fatality would eventually occur?
5. Edison Electric Institute — built for U.S. investor-owned electric utilities.
6. High-energy assessment plus identification of direct controls.
7. Missing, failed, weak, bypassed, degraded, unverified.
8. A model of layered organizational defenses, each with weaknesses ("holes"), that align only rarely to let harm through; developed by James Reason.
9. An active failure is the immediate, visible triggering action; a latent condition is the underlying systemic weakness that made it more likely.
10. Threats → Preventive Barriers → Top Event → Mitigative Barriers → Consequences.
11. 18 before (8 Core + 10 Supplemental); 9 after.
12. About 2% — small enough that IOGP folded excavation-related risk mostly into the Work Authorisation rule rather than keeping it as its own separate rule.

13. 1.8 metres (about 71 inches).
14. "Completed" describes an intended/claimed state; "verified" describes a confirmed, checked state — the gap between them is exactly where SIF potential often hides.
15. Retrieval-Augmented Generation — giving the model the actual Life-Saving Rule text/definitions at the moment it's tagging a report, so mapping is grounded in the real rules rather than the model's imprecise memory of them.
16. With rare-event classification, a model that always predicts "no" can score very high accuracy while missing every genuinely dangerous case.
17. The F2 metric, prioritizing recall over precision, because missing a real SIF-potential report is worse than an extra false alarm.
18. Any three of: VelocityEHS (AI PSIF Insights), Cority (Cortex AI, SIF scoring), Intelex (Input/Insight AI, predictive risk scoring).
19. Oil Industry Safety Directorate; established 1986.
20. OISD (under MoPNG) develops technical safety standards; PESO administers legal licensing under the Petroleum Act and Explosives Act, and has formally adopted around 11 OISD standards into its own mandatory rules.

Level 3 — Advanced (15 questions)

1. Explain why RiskRadar should not be "just an LLM call," using the specific architectural layers involved.
2. What does it mean for RiskRadar's scoring layer to be "auditable," and why does that matter more than raw model confidence?
3. Design (in words) a synthetic report that would test whether your barrier-failure extraction correctly handles a contradictory narrative.
4. Why should a report's Life-Saving Rule mapping allow multiple tags rather than forcing a single label?
5. Explain the difference between "prototype validation" and "real-world validation," and why blurring them is dangerous in a judge Q&A.
6. What does "precursor density" mean, and why is it a better ranking measure than raw report count?
7. Explain how the Swiss Cheese Model should change how RiskRadar's output is used downstream, specifically regarding blame.
8. Why does IOGP describe the Life-Saving Rules as "a final barrier," and what does that imply about what they are not?
9. What is the honest, defensible difference between RiskRadar and VelocityEHS's AI PSIF Insights?
10. Why does the domain model (Chapter 21) include a separate `process_safety_relevant` flag alongside the Life-Saving Rule tag?
11. Explain DEKRA's "last link in the chain" guidance and why it matters for extraction design.
12. What safeguard specifically addresses "automation bias," and how does it work mechanically?
13. Why is PR-AUC generally more informative than ROC-AUC for this specific classification problem?
14. Explain why OIL's HSE Transformation Plan (Phase-II) is a relevant fact for RiskRadar's pitch, beyond just "OIL cares about safety."
15. What would have to be true for NorthStar to honestly claim a specific accuracy percentage to a judge?

Level 3 Answer Key

1. A raw LLM call gives no evidence trail, can hallucinate hazards not in the text, and can't be audited or improved in a structured way when wrong; the hybrid architecture separates extraction (LLM/NLP) from scoring (a deterministic, auditable layer) so failures can be traced to a specific layer.
2. Auditable means every part of the score is traceable to specific extracted evidence from the source text; this matters more than raw confidence because a confident-sounding wrong answer is more dangerous than a well-evidenced uncertain one — confidence alone gives no way to check or correct the reasoning.
3. Example: a narrative stating "isolation completed" in one sentence, then later "isolation could not be confirmed by the second technician" — testing whether extraction correctly weighs the more specific, more operative, later statement over the earlier vaguer one.
4. Because real precursors often span more than one hazard/rule at once (e.g. the SIMOPS example touching both

Safe Mechanical Lifting and Hot Work); forcing a single tag would lose real information.

5. Prototype validation shows the pipeline behaves correctly on designed test cases; real-world validation shows actual performance on OIL's real operational language and base rates. Blurring them risks presenting synthetic-data performance as if it were a real-world accuracy claim — a credibility-destroying overclaim if a judge catches it.
6. Precursor density measures SIF-potential concentration relative to total report volume at a site/activity, not raw count — a small site with a high proportion of high-potential reports can be more urgent than a large site with many low-potential reports.
7. It means flagged reports should not automatically be read as "worker error" — a downstream investigation should still ask what latent organizational conditions (procedure design, staffing, time pressure) made the flagged active failure more likely.
8. It implies the Rules are not a replacement for the full management system, competent people, procedures, or training — they are the last line of defense an individual has personal control over once everything upstream has already failed.
9. VelocityEHS's public materials show general PSIF classification across several non-oil-and-gas verticals, without shown IOGP Life-Saving Rule mapping, oil-and-gas-specific hazard taxonomy, or a precursor-graph feature; RiskRadar is scoped specifically around those gaps.
10. Because a personal-safety event (e.g. a fall) and a process-safety event (e.g. a containment failure) can both map to the same Life-Saving Rule but carry very different escalation implications (process-safety events can affect far more people) — the flag lets downstream triage treat them differently even when the rule tag is shared.
11. DEKRA advises analyzing the last link in the causal chain that created the exposure, not the entire causal history at once — for extraction, this means prioritizing the most proximate barrier failure described in the text rather than trying to reconstruct a full multi-step causal narrative automatically.
12. Evidence display — showing the specific extracted fields and source text behind every score — is designed to keep a human reviewer actively engaged in checking the reasoning, rather than passively trusting a bare number.
13. Because the dataset is heavily imbalanced (SIF-potential reports are a minority), ROC-AUC can look artificially strong even when precision on the positive class is weak; PR-AUC focuses specifically on performance on the minority (positive) class, which is the class that actually matters here.
14. It shows OIL is, right now, actively investing in strengthening its own HSE Management System — meaning RiskRadar arrives at a moment where a credible complementary tool has an organizationally receptive audience, not a cold pitch into an indifferent system.
15. NorthStar would need real, authenticated, labeled OIL historical report data, a proper held-out test set from that real data, and metrics computed and reported using standard, disclosed methodology (with the recall/precision tradeoff made explicit) — none of which is available at hackathon-prototype stage.

Level 4 — Judge Simulation (10 questions)

Answer these out loud, to a teammate, without notes, in under 90 seconds each. If you can't, go back and reread the relevant chapter.

1. "Convince me this isn't just a dashboard wrapped around ChatGPT."
2. "Your competitor already sells this. Why does OIL need you?"
3. "Show me the real OIL data proving your model works."
4. "What happens the one time your AI is wrong about a report that turns out to matter?"
5. "Why should I trust a hackathon team's safety judgment over trained HSE professionals'?"
6. "What's the actual dollar value of this to OIL?"
7. "Walk me through exactly what happens, technically, from the moment a report is submitted to the moment it appears on your priority queue."
8. "If I gave you OIL's real data tomorrow, what would you actually do with it?"
9. "What's the single biggest weakness in your project, in your own words?"
10. "In one sentence, why should this win?"

(There is no single "correct" answer key for Level 4 — grade yourselves against how directly and honestly each answer draws on Chapters 25, 33, 28, 32, 39, 36, 42, and the Final Mental Model below.)

Chapter 44 — Cheat Sheets

SIF Cheat Sheet

- SIF = actual serious injury or fatality. SIF Potential = could reasonably have been one.
- Test: if repeated dozens/hundreds of times, would it eventually be serious? (DEKRA)
- ~25% of OSHA recordables carry real SIF exposure potential (DEKRA).
- Fatality rates flat for ~20 years even as minor-injury rates fell (DEKRA).
- Precursor = a high-risk situation where controls are absent, ineffective, or not complied with (DEKRA).

HSE Cheat Sheet

- HSE = Health, Safety, Environment. HSSE adds Security.
- Workflow: Observation → Report → Review → Classification → Risk Assessment → Investigation → Corrective Action → Verification → Closure → Trend Analysis.
- RiskRadar sits right after Report, before Review — and makes Trend Analysis continuous.

OIL Cheat Sheet

- Founded 1959; Maharatna since Aug 2023; Ministry of Petroleum & Natural Gas.
- HQ: Duliajan, Assam (field/registered); corporate office in Noida.
- Core: E&P of crude oil and natural gas, LPG, pipelines (Naharkatiya–Barauni); NRL refining subsidiary since 2021; expanding into solar/green hydrogen/CCUS.
- FY24–25 best-ever LTIFR: 0.071. Confirmed practices: HAZOP, QRA, JSA, near-miss portals, KAVACH initiative, active HSE Transformation Plan Phase-II.

IOGP Life-Saving Rules Cheat Sheet

Bypassing Safety Controls · Confined Space · Driving · Energy Isolation · Hot Work · Line of Fire · Safe Mechanical Lifting · Work Authorisation · Working at Height. 9 rules since 2017–18 (was 18). 376 lives potentially saved 2008–2017 by following them.

Oil & Gas Hazards Cheat Sheet

Stored/pressurized energy · electrical energy · loss of containment · H_2S/toxic gas · fire/explosion · working at height · dropped objects/suspended loads · line of fire · confined space · hot work · vehicle movement · lifting operations · authorization gaps · bypassed controls · excavation · process upset/blowout.

Barrier Cheat Sheet

Types: physical/engineering, procedural, human, administrative, preventive, mitigative. Failure modes: missing, failed, weak, bypassed, degraded, unverified.

AI/NLP Cheat Sheet

- Keyword matching fails on negation/context (“isolation completed” vs “not verified”).
- Hybrid architecture: LLM/NLP extraction → domain taxonomy → auditable scoring → human review.
- Prioritize recall over precision (F2 metric) — validated by real PSIF research (PMC10998882).
- RAG grounds Life-Saving Rule mapping in the real rule text, reducing hallucination risk.

RiskRadar Architecture Cheat Sheet

```
Report → NLP Preprocessing → AI Extraction (hazard, activity, barrier,
exposure) → Domain Taxonomy Mapping → Scoring/Rule Engine → SIF
Assessment + Evidence + Confidence → Human Review (low confidence /
high consequence) → Dashboard (priority queue, explanation, patterns)
```

Judge Q&A Cheat Sheet (fastest-recall version)

- “Why AI, not keywords?” → isolation completed vs. not verified.
- “What if it’s wrong?” → confidence thresholds, evidence display, mandatory human review for high-consequence cases.
- “Where’s your data?” → synthetic, honestly labeled as such; architecture ready for real data substitution.
- “Who else does this?” → VelocityEHS, Cority, Intelex — we know, and we’re scoped around the gaps (IOGP mapping, precursor chain, OIL-specific).
- “Prevents fatalities?” → No — surfaces precursors and prioritizes intervention. Never say “prevents.”

Chapter 45 — Master Glossary (A–Z)

(Terms already given full treatment in Chapter 7 are listed here only in short form, to avoid duplication — go to Chapter 7 for the complete explanation.)

- **ALARP** — As Low As Reasonably Practicable; a risk-reduction sufficiency principle.
- **Barrier** — see Chapter 7/11.
- **BERT** — Bidirectional Encoder Representations from Transformers; a transformer model well suited to text classification.
- **Bow-Tie** — a diagram connecting threats, preventive/mitigative barriers, a top event, and consequences (Ch.13).
- **CAPA** — Corrective and Preventive Action (Ch.7).
- **Confusion Matrix** — a table of predicted vs. actual classification outcomes (Ch.31).
- **DEKRA** — a global safety consultancy and originator of much of the SIF-prevention field (Ch.10).
- **DGMS** — Directorate General of Mines Safety, India.
- **EEI** — Edison Electric Institute; U.S. electric-utility trade body behind the SCL Model (Ch.10).
- **Embeddings** — numeric vector representations of text capturing meaning (Ch.23).
- **F1 / F2 score** — metrics balancing precision and recall, with F2 weighting recall more heavily (Ch.31).
- **HAZID / HAZOP** — Hazard Identification / Hazard and Operability Study (Ch.7, Ch.17).
- **HIRA** — Hazard Identification and Risk Assessment (Ch.17).
- **HIPO** — High Potential Event (Ch.7).
- **Hybrid AI** — an architecture combining AI extraction with deterministic/auditable scoring logic (Ch.25).
- **ICAM** — Incident Cause Analysis Method (Ch.19).
- **IOGP** — International Association of Oil & Gas Producers, source of the Life-Saving Rules (Ch.15).
- **ISO 45001 / 14001 / 31000** — international standards for occupational health & safety, environmental management, and risk management respectively (Ch.35).
- **JSA / JHA** — Job Safety Analysis / Job Hazard Analysis (Ch.7, Ch.17).
- **KAVACH** — a named OIL safety initiative aimed at improving safety practices (Ch.5) — exact internal mechanics not publicly detailed.
- **Latent Condition** — an underlying systemic weakness, per the Swiss Cheese Model (Ch.12).
- **Life-Saving Rule (LSR)** — one of IOGP's nine defined rules (Ch.15).
- **LLM** — Large Language Model (Ch.24).
- **LOTO** — Lockout/Tagout (Ch.7, Ch.18).
- **LTIFR** — Lost Time Injury Frequency Rate (Ch.7).
- **Maharatna** — the highest category of Indian Central Public Sector Enterprise, granted to OIL in August 2023 (Ch.2).
- **MOC** — Management of Change (Ch.7).
- **MoPNG** — Ministry of Petroleum and Natural Gas, Government of India.
- **NRL** — Numaligarh Refinery Limited, OIL's refining subsidiary since March 2021 (Ch.2).
- **OISD** — Oil Industry Safety Directorate, India (Ch.35).
- **PESO** — Petroleum and Explosives Safety Organisation, India (Ch.35).
- **PNGRB** — Petroleum and Natural Gas Regulatory Board, India (Ch.35).
- **PR-AUC / ROC-AUC** — Precision-Recall / Receiver Operating Characteristic Area Under Curve metrics (Ch.31).
- **Precursor** — see Chapter 9.
- **PSIF** — Potential Serious Injury or Fatality; the term used by EEI's SCL Model and by several commercial products/research papers, closely related to SIF Potential (Ch.10, Ch.33–34).
- **PTW** — Permit to Work (Ch.7, Ch.18).
- **RAG** — Retrieval-Augmented Generation (Ch.24).
- **RCA** — Root Cause Analysis (Ch.7, Ch.19).

- **SCL Model** — EEI’s Safety Classification and Learning Model (Ch.10).
- **SIF / SIFp / SIF Exposure** — see Chapter 8.
- **SIMOPS** — Simultaneous Operations (Ch.7).
- **Start Work Checks** — IOGP’s pre-task verification tool complementing the Life-Saving Rules (Ch.15).
- **Swiss Cheese Model** — James Reason’s layered-defense model (Ch.12).
- **TapRooT®** — a proprietary root-cause-analysis methodology and software (Ch.19).
- **Tripod Beta** — a root-cause methodology originating in oil & gas (Ch.19).
- **TRIR** — Total Recordable Incident Rate (Ch.7).
- **VelocityEHS** — vendor of “AI PSIF Insights,” a directly comparable commercial product (Ch.33).
- **XGBoost** — a gradient-boosting machine-learning algorithm, used in the PMC10998882 PSIF-classification research (Ch.34).

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NorthStar’s Final Mental Model

```
OIL has large volumes of safety reports
              ↓
Reports contain hidden information about dangerous situations
              ↓
Some low-severity events have high fatal potential (DEKRA-verified)
              ↓
Manual, periodic review can miss recurring patterns
              ↓
RiskRadar understands the report (NLP extraction)
              ↓
Detects SIF potential (hybrid, auditable scoring)
              ↓
Identifies hazard + exposure + barrier failure (explainable evidence)
              ↓
Maps the relevant IOGP Life-Saving Rule
              ↓
Finds recurring patterns across sites and activities
              ↓
Ranks sites / activities / hazards by precursor density
              ↓
HSE team investigates and acts — RiskRadar assists, never replaces
              ↓
Risk is addressed before it escalates into a real SIF
```

If you remember only 20 things

1. SIF potential ≠ actual severity. This is the whole project.
2. DEKRA’s test: if repeated many times, would it reasonably end badly?
3. ~25% of recordables carry real SIF potential — most don’t.
4. A precursor is a high-risk situation where a control is absent, ineffective, or not complied with.
5. “Isolation completed” ≠ “isolation verified” — this single distinction is your best concrete example.
6. There are 9 IOGP Life-Saving Rules today, down from 18 since 2017–18.
7. The Rules are a final personal barrier, not a replacement for the full management system.
8. Process safety (containment, major events) ≠ personal safety (individual, acute harm).
9. OIL is Maharatna-status, HQ’d in Duliajan, Assam; FY24–25 best-ever LTIFR of 0.071; actively mid-way through an HSE Transformation Plan.
10. RiskRadar should never be “just an LLM call” — extraction and scoring must be separate, auditable layers.
11. Recall matters more than precision here — a real 2024 peer-reviewed study on this exact problem validates that.
12. You do not have OIL’s real data — say so honestly, and show a well-designed synthetic dataset instead.
13. VelocityEHS, Cority, and Intelex already ship AI SIF/PSIF features — know this, and lead with your honest differentiation.
14. Never claim RiskRadar “prevents” or “predicts” a fatality — it “surfaces,” “prioritizes,” and “supports.”
15. Every score needs visible evidence — this is what makes the system explainable, auditable, and trustworthy.
16. Precursor density (concentration, not raw count) is the right way to rank sites and activities.
17. The Swiss Cheese Model means a flagged report is a starting point for investigation, not an automatic verdict on a worker.
18. Build the per-report SIF Precursor Chain before the aggregate dashboard charts — that’s your strongest demo asset.
19. Accuracy alone is a dangerous metric for a rare-event problem — know precision, recall, F1, and PR-AUC.
20. Being honest about weaknesses in front of judges is a strength, not a liability.

NorthStar’s 30-Second Explanation

"OIL collects thousands of safety reports, but a report's actual severity and its fatal potential aren't the same thing — a near-miss with no injury can be more dangerous than an incident with a minor one. RiskRadar reads OIL's free-text safety reports, identifies the ones that quietly show serious-injury-or-fatality potential, maps them to the relevant IOGP Life-Saving Rule, and shows HSE teams where the same precursor is recurring across sites — so they can act before it escalates into a real fatality, not after."

NorthStar's 2-Minute Explanation

"Every safety-critical company generates two kinds of information: what actually happened, and what could have happened. Research from organizations like DEKRA shows these are driven by different causes — about a quarter of typical recordable incidents carry real serious-injury-or-fatality potential, and that potential doesn't track cleanly with how bad the actual outcome was. A near-miss where nobody was hurt can carry more fatal potential than an incident that produced a minor injury — because what matters is what could have happened if the timing or circumstances had been slightly worse.

OIL's HSSE platform already collects Unsafe Act, Unsafe Condition, near-miss, and incident reports at scale — but review today happens manually and periodically, which is well-suited to counting what happened, and poorly suited to catching a quiet, recurring precursor pattern before it repeats one time too many.

RiskRadar is an AI/NLP system that reads that free text, and instead of matching keywords — which fails on distinctions like 'isolation completed' versus 'isolation not verified' — it uses a hybrid architecture: an AI extraction layer pulls out the hazard, the activity, the exposure, and specifically which safety barrier failed, and a separate, auditable scoring layer turns that structured evidence into a SIF-potential score with visible reasoning, not a black-box number. High-potential and low-confidence cases are always routed to a human — RiskRadar is built to prioritize and explain, not to decide alone.

Each flagged report is also mapped to the relevant IOGP Life-Saving Rule, and checked against past reports to see if the same precursor — say, isolation-verification failures during maintenance — is recurring across multiple sites, which is exactly the kind of pattern a monthly manual review is structurally likely to miss.

We know this isn't an entirely unclaimed space — commercial products like VelocityEHS's AI PSIF Insights already do AI-based potential-severity classification, and our own honest positioning is that RiskRadar's contribution is the IOGP-aligned, oil-and-gas-specific, explainable version of that idea, purpose-built for how OIL actually works — not a claim that nobody has ever thought of this before.

We don't have access to OIL's real report data, and we say that plainly rather than pretend otherwise — our prototype runs on a carefully designed synthetic dataset, built specifically to include the hard cases: ambiguous reports, multi-hazard reports, and reports where the language contradicts itself. The architecture is built so real OIL data could be substituted in directly.

The goal isn't to predict a specific fatality — that's not a claim we'd ever make. It's to move HSE attention from counting what already happened toward acting on what's quietly building toward happening — before it does."